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ORIGINAL PAPER

# Effects of La and Ce treatment on changes in inclusion characteristics of 304 stainless steel during industrial production

Zhan-hua Shen[1,2] · Lei Huang[3] · Hua Zhang[1,2] · Xue-zhong Huang[3] · Cheng-song Liu[1,2] · Yu-zhong Yang[1,2] · Hong-wei Ni[1,2]

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## Abstract

The extent of inclusion modification achieved after adding various contents of rare earth elements to 304 stainless steel in practical applications was explored, additionally the pitting behavior of the inclusions in a chloride environment before and after modification. The mechanisms by which the compositions of inclusions evolved throughout the production process in response to various rare earth element (REE) contents were investigated, and their consequential impact on the corrosion resistance of steel was evaluated. The results indicated that with increasing rare earth content from 0 to $50.37 \times 10^{-6}$, REE inclusions were preferentially generated in the following order when they were modified: $REAlO_3 \rightarrow RE_2O_2S \rightarrow RE_2O_3$. The inclusion size and density were negatively correlated and positively correlated with the REE content in 304 stainless steel, respectively. The formation of Mg–Al spinel and MnS in the continuously cast billet was suppressed by the addition of rare earth elements. The voltage potential differences between the REE-modified inclusions and the steel substrate were obviously less than those between the unmodified inclusions and the steel substrate. The addition of rare earth elements contributed to the improvement in the corrosion resistance of 304 stainless steel.

**Keywords** 304 stainless steel · Modification of inclusion · Rare earth treatment · Corrosion resistance · Industrial testing

## 1 Introduction

Stainless steel is commonly employed in a variety of maritime applications, including ships, marine oil and gas extraction and transportation systems, offshore platforms, and other related fields. Compared with the terrestrial environment, ocean currents change drastically with the season. Stainless steel is subject to the low-frequency reciprocating stress and impact force generated by waves, tides and currents, and its service environment is extremely harsh. The key considerations of its physical properties include toughness, strength, plasticity, welding performance, corrosion resistance, and uniformity [1–8].

Rare earth elements (REEs) are highly important in the metallurgical industry and are considered special additions to improve the properties and quality of steels. Rare earth elements can exist in steel in three primary forms. The first form is the uniform dissolution of rare earth elements in the steel solution in the solid state, similar to other conventional alloying elements. The second form involves rare earth elements acting as reactive elements, which are combined with the sulfur and oxygen in steel to generate nonmetallic compounds. The third form involves the formation of compounds with manganese and vanadium in steel [9, 10]. These compounds precipitate as nonuniform nucleation cores before the steel solidifies. For steel, nonuniform nucleation reduces the degree of subcooling during crystallization, refines the solidification structure,

✉ Hua Zhang
huazhang@wust.edu.cn

✉ Cheng-song Liu
liuchengsong@wust.edu.cn

[1] Key Laboratory for Ferrous Metallurgy and Resources Utilization of Ministry of Education, Wuhan University of Science and Technology, Wuhan 430081, Hubei, China

[2] The State Key Laboratory of Refractories and Metallurgy, Wuhan University of Science and Technology, Wuhan 430081, Hubei, China

[3] Guangxi Key Laboratory of Special Steel and New Materials, Guangxi BG New Materials Co., Ltd., Beihai 536000, Guangxi, China

and mitigates segregation. Appropriate addition of rare earth elements contributes to improvements in the impact toughness, plasticity, hardness, corrosion resistance, and fatigue toughness of steel by optimizing the microstructure and inclusions [11–14].

In recent years, many scholars have conducted extensive studies on the effects of rare earth treatments on steel. Meng et al. [15] produced high manganese steel containing rare-earth yttrium using a vacuum induction furnace, and the composite inclusions in the high manganese steel were analysed in a multidimensional structure. After modification with rare-earth yttrium, the $MnS–Al_2O_3$ composite inclusions in the steel were transformed into $Y_2S_3–Y_2O_2S$ composite inclusions with smaller size and near-spherical shape. In laboratory experiments conducted by Zhang et al. [16], researchers examined the transient changes in inclusions in Si–Mn stainless steel with the addition of $40 \times 10^{-6}$, $100 \times 10^{-6}$, and $400 \times 10^{-6}$ cerium. As the cerium content increased, the inclusions changed from having a composition of $Al_2O_3–SiO_2–MnO–CaO$ to $Ce_2O_3–Al_2O_3–SiO_2–MnO–CaO$ to $Ce_2O_3–CeS$. Luo et al. [17] proposed that the evolution of $Al_2O_3$ inclusions formed by Al deoxidation in 20CrNiMo gear steel followed the route of $Al_2O_3 \rightarrow CeAlO_3 \rightarrow Ce_2O_2S \rightarrow Ce_2O_2S + CeS$. Zhang et al. [18] examined the impacts of Ce refining agents on the solidification structure and mechanical properties of ferritic stainless steel and discussed the associated mechanism. The results showed that the addition of 0.011 wt.% Ce and 0.023 wt.% Ce refining agents substantially enhanced the ductility and toughness of the material. Kim et al. [19] studied the impacts of rare earth additions on the pitting resistance characteristics of super duplex stainless steels. The addition of rare earth elements to the base alloy led to the formation of (Mn, Cr, Si, Al, Ce) oxides and (Mn, Cr, Si, Ce) oxides, enhancing the resistance to pitting and diminishing the favored interfacial region for pitting initiation. Wang et al. [20] characterized the pitting corrosion behavior of 13Cr4Ni martensitic stainless steel (BASE) and rare-earth modified stainless steel (REM) in a 0.1 mol/L NaCl solution. The results showed that REM steel was more susceptible to inducing metastable pitting nucleation and repassivation than BASE steel was, and REM steel inhibited the transition from metastable pitting to stable pitting.

Considering the persistent challenge of fully eliminating inclusions from steel, it has become imperative to explore methodologies aiming at mitigating the detrimental effects of inclusions. In this study, the focus was modifying inclusions by incorporating rare earth elements into 304 stainless steel, which is widely used in marine industries. This strategy effectively combated corrosion induced by inclusions, particularly pitting corrosion, while concurrently increasing the resistance of steel to intergranular corrosion [21, 22]. In addition, the extent of inclusion modification achieved after adding various contents of rare earth elements in practical applications was also explored. Furthermore, the mechanisms by which the compositions of inclusions evolved throughout the production processes of 304 stainless steel in response to various REE contents were investigated, and their consequential impact on the corrosion resistance of steel was also evaluated.

## 2 Experimental methods

Four heats of 304 stainless steel containing low-phosphorus iron and composite microalloying elements were subjected to full-scale industrial production processing. This processes included smelting iron in a high frequency induction furnace to produce about 90 t of crude molten steel, refining the crude molten steel in an argon-oxygen decarburization (AOD) converter, treating the refined steel in a ladle furnace (LF), and then processing the steel through continuous casting, slab heating, rolling, solid solution treatment and acid pickling (Fig. 1).

During the LF refining process, pure La and Ce rare earth metals were packed together in a 304 stainless steel container and then added into the molten steel. Counterweights hung below the container to facilitate the submergence of the rare earth elements into the molten steel. The amounts of rare earth elements added to each heat were as follows: no rare earth elements were added to heat 1; 6 kg of rare earth elements were added to heat 2; 12 kg of rare earth elements were added to heat 3; and 18 kg of rare earth elements were added to heat 4. When rare earth elements were added during alloying, the ratio of Ce to La was 1:1. Calcium wire was also introduced into the steel at the appropriate temperature during LF refining, ensuring that the alloy composition and sulfur content met the specified criteria. The treatment involved feeding 1 m of Ca28Si60 core wire (with a core diameter of $\phi$13mm, a content of 28 wt.% calcium, and a core powder quality of 210 g/m) per tonne of steel at a feeding rate of at least 60 m/min. The detailed chemical composition of the silicon–calcium wire is shown in Table 1. Rare earth elements were added after temperature modification and composition fine-tuning. There was a minimum soft blowing time of 3 min after addition.

Slag and steel samples were taken from each heat upon entry into the LF and before exiting the LF. A NexION 1000Gx Inductively Coupled Atomic Emission Spectrometer was employed to ascertain the concentrations of elements including Al, Si, Ca, Ce, and La in the steel, and the oxide components in the slag were determined by X-ray fluorescence spectrometer. A Leco TC500 N/O analyser was utilised to detect the oxygen content in the steel. An

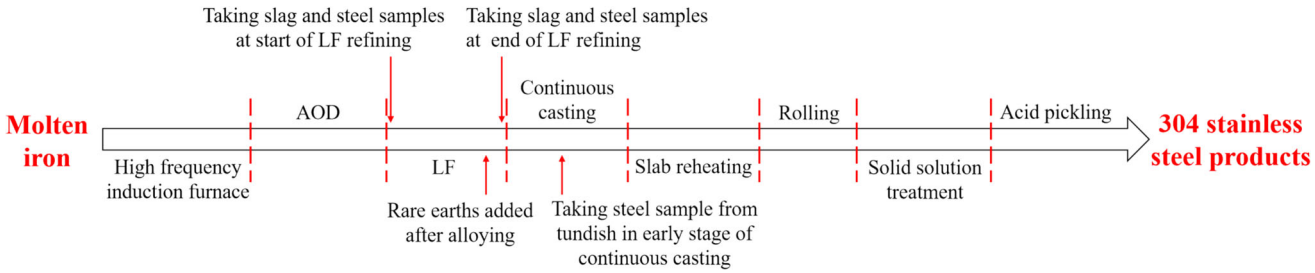



**Fig. 1** Industrial production processes of 304 stainless steel

**Table 1** Main chemical compositions of cored wire

| Element | Ca | Si | Mg + Ba | B + Zr + Ti |
|---|---|---|---|---|
| Content/wt.% | 28 | 60 | 2–8 | 4–10 |

Infrared Carbon Sulphur Analyser was employed to determine the sulphur content in the steel. 10 mm × 10 mm metallographic specimens were prepared from the original samples and subsequently ground and polished in order to facilitate the analysis of the morphology and composition of typical inclusions under a scanning electron microscope. Furthermore, an automated inclusions scanning analyser (Aspex) was employed to determine the number, composition and size of inclusions existing in the steel samples obtained from the various production processes. An area size of around 17 $mm^2$ was selected for measurement, which was deemed to be sufficiently representative for evaluating the inclusion characteristics.

The kinetic potentiodynamic polarization curves and electrochemical impedance of the samples of continuous casting billets were analyzed via an electrochemical workstation's three-electrode system. The reference electrode used was a saturated calomel electrode, and the auxiliary electrode was a platinum electrode. The samples were wrapped externally with insulating glue, exposing only a 10 mm × 10 mm test surface. The medium solution was a 3.5 wt.% NaCl solution. The open-circuit potential in solution was determined for 3600 s at the beginning of the electrochemical measurement. Dynamic potentiodynamic polarization was measured from −1.2 to 1.8 V with a potential scanning rate of 0.0002 V $s^{-1}$. The electrochemical impedance was tested over a frequency range of 0.01–1 × $10^5$ Hz using an AC excitation signal amplitude of 0.005 V. Scanning Kelvin probe force microscopy (SKPFM) was used to observe the morphologies of typical inclusions and measure the potential difference between them and the steel matrix.

In the early stage of the corrosion process, the immersion test is a simple and reliable tool for analysing localised corrosion mechanisms. The pitting mechanism of inclusions of each heat of continuous casting billets at room temperature was studied via immersion tests for 50 h in 10 vol.% $FeCl_3$ solution. All tests were conducted at least three times to ensure reproducible results. The methods for the immersion treatment experiments, including solution preparation, frequency of solution renewal and sample handling procedures, were as follows: (1) Solution preparation method: 0.37 g of ferric chloride hexahydrate was dissolved into deionized water, and the solution volume was adjusted to 100 mL; (2) The frequency of solution renewal: after 10 h of immersion treatment, the original solution was replaced with new solution; (3) Sample handling procedure: after the immersion process, the samples were subsequently ground with 400, 800, 1200, 1500, and 2000 grits of sandpapers, and then polished with a 1 μm diamond paste until there were no scratches. At the end of etching and polishing, the sample surface was rinsed clean using alcohol solution.

## 3 Results

### 3.1 Changes in inclusion characteristics during industrial production processes

#### 3.1.1 LF refining process

Table 2 displays the chemical composition of steel in each heat after the LF process. Table 3 shows the composition of the slag during each LF heat. In Table 3, serial number 1 after the heat number indicates the composition of the slag at the onset of LF refining of the heat, whereas serial number 3 signifies the slag composition after LF refining of the heat. The rare earth content in the steel indicated a significantly greater yield of Ce than that of La.

Figure 2 shows the typical inclusions in the steels after LF refinement. Figure 2a shows the typical CaO–$SiO_2$–$Al_2O_3$–MgO inclusion in the steel sample without the addition of rare earth elements. The CaO–$SiO_2$–$Al_2O_3$–MgO inclusions were generally homogeneous and spherical. Figure 2b–d shows the inclusions in the steel samples with REE contents of 2.2 × $10^{-6}$, 25.2 × $10^{-6}$, and

**Table 2** Chemical composition of steel in each heat after LF process (wt.%)

| Element | Cr | Ni | Mn | Si | Mg | Al | Ca | S | TO | Ce | La |
|---|---|---|---|---|---|---|---|---|---|---|---|
| LF-1 | 18.121 | 8.118 | 0.783 | 0.280 | $29 \times 10^{-4}$ | $9.9 \times 10^{-4}$ | $22 \times 10^{-4}$ | $11.1 \times 10^{-4}$ | $49 \times 10^{-4}$ | 0 | 0 |
| LF-2 | 18.090 | 8.023 | 0.746 | 0.190 | $23 \times 10^{-4}$ | $11 \times 10^{-4}$ | $29 \times 10^{-4}$ | $21.1 \times 10^{-4}$ | $67.5 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | $1 \times 10^{-4}$ |
| LF-3 | 18.176 | 8.109 | 0.790 | 0.210 | $27 \times 10^{-4}$ | $6.4 \times 10^{-4}$ | $23 \times 10^{-4}$ | $19.8 \times 10^{-4}$ | $53 \times 10^{-4}$ | $16 \times 10^{-4}$ | $9.2 \times 10^{-4}$ |
| LF-4 | 18.231 | 8.096 | 0.754 | 0.296 | $39 \times 10^{-4}$ | $9.4 \times 10^{-4}$ | $24 \times 10^{-4}$ | $20 \times 10^{-4}$ | $47 \times 10^{-4}$ | $32.1 \times 10^{-4}$ | $18.3 \times 10^{-4}$ |

**Table 3** Slag compositions at different stages of each heat (wt.%)

| Serial number | Content | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | $SiO_2$ | $Al_2O_3$ | CaO | MgO | $Cr_2O_3$ | MnO | Ni | $Fe_2O_3$ | LOI |
| LF-1–1 | 26.50 | 1.90 | 61.72 | 4.31 | 0.095 | 0.550 | 0.0200 | 0.120 | 0.20 |
| LF-1–3 | 24.05 | 1.96 | 63.56 | 4.21 | 0.110 | 0.018 | 0.0025 | 0.180 | 0.04 |
| LF-2–1 | 29.16 | 1.70 | 59.59 | 3.87 | 0.140 | 0.065 | 0.0022 | 0.160 | 1.93 |
| LF-2–3 | 29.87 | 2.10 | 58.12 | 4.43 | 0.110 | 0.059 | 0.0330 | 0.140 | 0.32 |
| LF-3–1 | 27.96 | 1.75 | 58.37 | 5.91 | 0.085 | 0.031 | 0.0020 | 0.110 | 0.45 |
| LF-3–3 | 29.26 | 1.75 | 56.67 | 5.97 | 0.067 | 0.028 | 0.0020 | 0.085 | 0.42 |
| LF-4–1 | 30.92 | 1.99 | 57.08 | 4.67 | 0.150 | 0.075 | 0.0010 | 0.078 | 0.09 |
| LF-4–3 | 30.42 | 2.41 | 56.61 | 5.05 | 0.150 | 0.060 | 0.0010 | 0.110 | 0.18 |

LOI—Loss on ignition

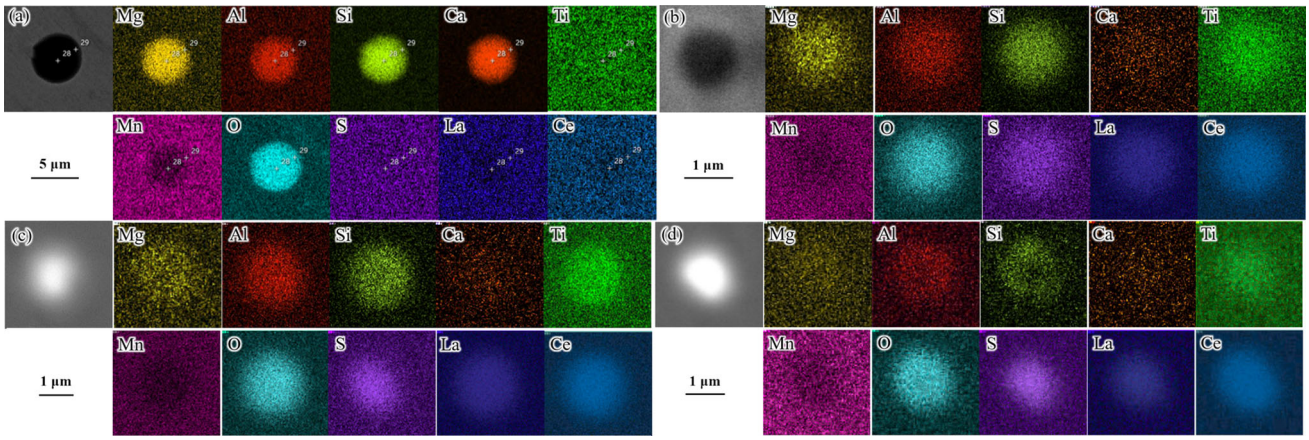



**Fig. 2** Morphologies and elemental distributions of typical inclusions in samples of 304 stainless steel after LF refining. **a** LF-1; **b** LF-2; **c** LF-3; **d** LF-4

$50.37 \times 10^{-6}$, respectively. The main inclusions in those steel samples were all generally composed of Mg–Al–Si–Ca–Ti–La–Ce–O–S, which were uniform in composition and irregular in shape. In those samples, most of the inclusions with high REE contents presented white by the scanning electron microscopy (SEM) in backscattered electron (BSE) mode.

Figure 3 shows the results of SEM automated analyses of the LF steel samples. Figure 3a shows the average diameters and number densities of the inclusions. From heat 1 to heat 4, as the rare earth element content in the LF samples increased from 0 to $2.2 \times 10^{-6}$ to $25.2 \times 10^{-6}$ and to $50.37 \times 10^{-6}$, the number density of inclusions in the LF samples first increased but then slightly decreased, from 11.28 to 76.60 to 72.18 and to 62.24 $mm^{-2}$, whereas the average diameter of inclusions in the LF samples gradually decreased from 2.12 to 1.37 to 1.33 μm and then to 1.28 μm. Figure 3b shows the distributions of inclusions of different sizes. Without the addition of rare earth elements, approximately 87% of the total inclusions have

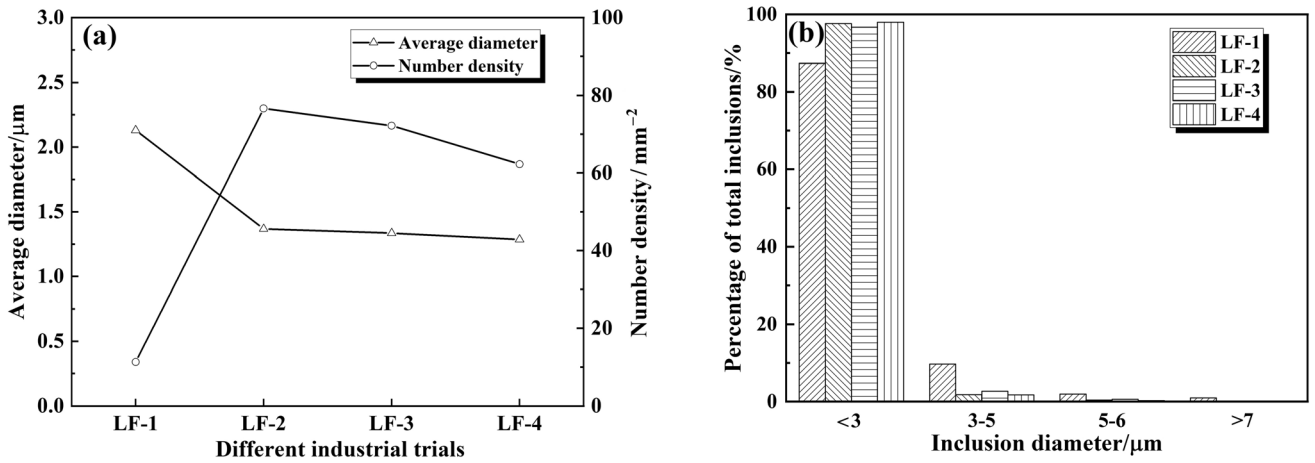



**Fig. 3** Changes in average diameter and number density (**a**) and size distribution (**b**) of total inclusions in 304 stainless steels after performing different LF refinement processes

diameters of < 3 μm, and approximately 10% of the total inclusions have diameters in the range of 3–5 μm. After rare earth elements were added, more than 95% of the total inclusions had diameters of < 3 μm. The addition of rare earth elements was obviously conducive to the refinement of the inclusions.

Figure 4 displays the compositional distributions of the inclusions in the steel samples after the LF refining process. Figure 4a shows the compositional distribution of inclusions in LF-1, indicating that without the addition of rare earth elements, the main inclusions in the steels after LF refining were generally complex CaO–$SiO_2$–$Al_2O_3$–MgO oxides. Figure 4b–d shows the compositional distributions

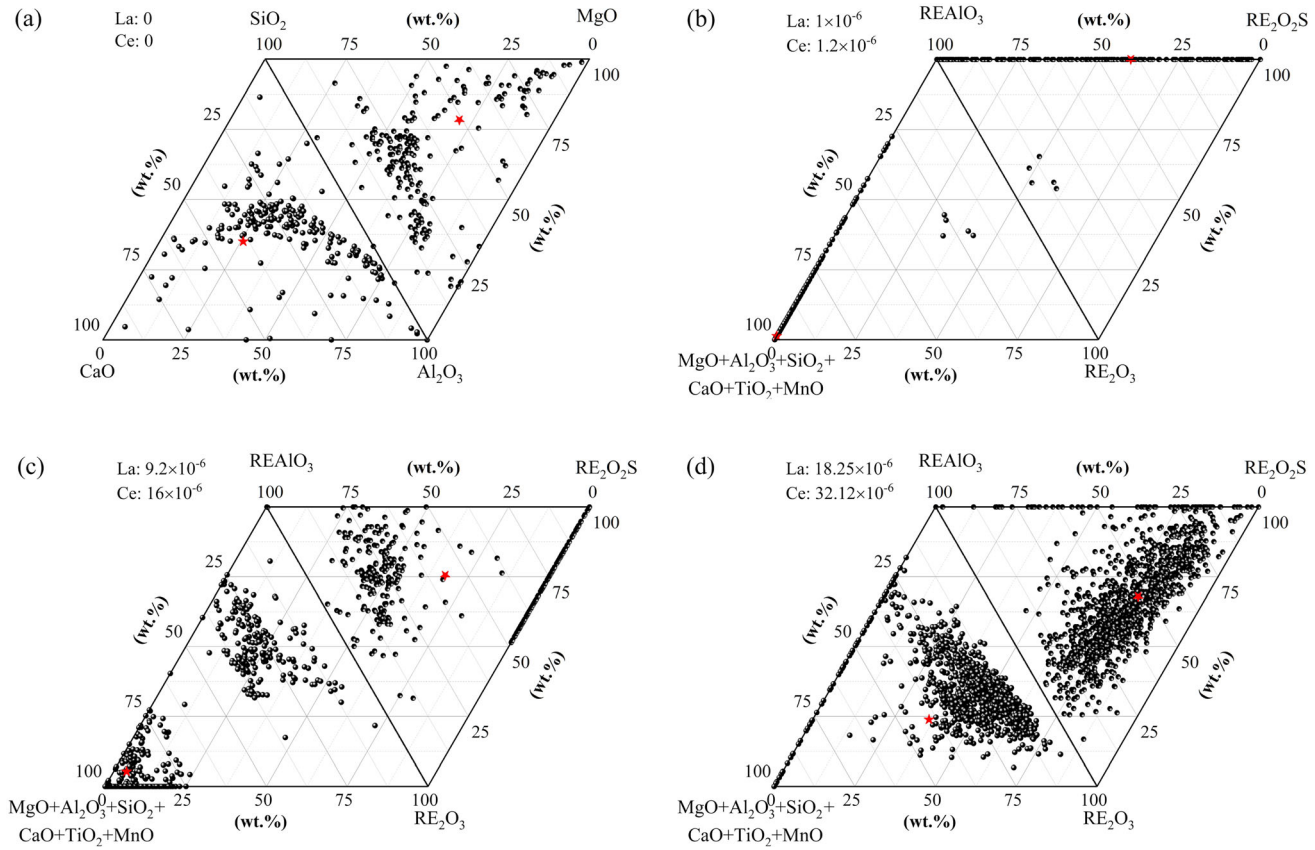



**Fig. 4** Distribution of inclusion compositions at end of LF refining process. **a** LF-1; **b** LF-2; **c** LF-3; **d** LF-4

of inclusions in LF-2, LF-3, and LF-4, respectively. Since after the treatment by Ce and La rare earth elements, the inclusions generally presented homogeneous morphology as shown in Fig. 2, the contents of $REAlO_3$ and $RE_2O_2S$ were determined by direct conversion from the measured concentrations of Al and S in the inclusions, respectively. The remained rare earth contents bound in the inclusions were regarded as the $RE_2O_3$ oxide. The results showed that as the content of rare earth elements (Ce + La) increased from 0 to $2.2 \times 10^{-6}$, the main inclusions in the steel after LF refining gradually transformed from the $CaO–SiO_2–Al_2O_3–MgO$ phase to the $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$ and $RE_2O_2S$) phases. As the content of rare earth elements (Ce + La) increased to $25.2 \times 10^{-6}$ in the LF-3 steel sample, the main inclusions evolved into $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases, with the content of $RE_2O_3$ being the lowest among these components. When the content of rare earth elements (Ce + La) reached $50.37 \times 10^{-6}$ in the LF-4 steel sample, the main inclusions were still $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases, with the content of $REAlO_3$ being the lowest among these components. According to the results shown in Fig. 4, the generation order of rare earth inclusions was $REAlO_3 \rightarrow RE_2O_2S \rightarrow RE_2O_3$ with increasing rare earth content from 0 to $50.37 \times 10^{-6}$. During inclusion modification, $REAlO_3$ and $RE_2O_2S$ inclusions initially formed. With increasing REE content, the amount of the $REAlO_3$ phase gradually decreased, whereas the contents of the $RE_2O_2S$ and $RE_2O_3$ phases in the inclusions increased. Figure 5 shows the average compositions of the total inclusions in the steel after the LF refining process. From heat 1 to 4, as the rare earth content in the LF samples increased from 0 to $2.2 \times 10^{-6}$ to $25.2 \times 10^{-6}$ and to $50.37 \times 10^{-6}$, the average rare earth inclusions content of the inclusions in the LF samples gradually increased from 0 to 3.01% to 15.10% and to 73.44%. Increasing the REE content in the steel obviously resulted in an increase in the rare earth content of the inclusions.

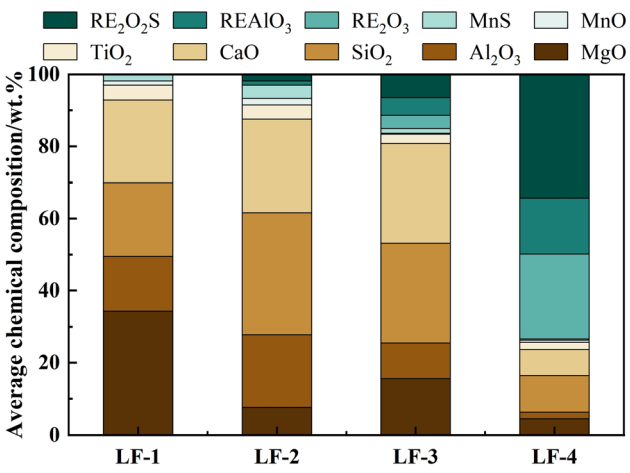



**Fig. 5** Average compositions of inclusions after LF refining

### 3.1.2 Continuous casting process

During the cooling process of the continuously cast billets, the cooling rate of the outer layer is greater than that of the inner layer. The cooling rate of the continuously cast billet exerts influence on the morphologies and composition distributions of inclusions. Consequently, the steel samples were machined from the surface and the centre of the continuous casting billet for analyzing the characteristics of inclusions in them.

Figure 6 shows the number densities and size distributions of the inclusions in the continuously cast billets. Hereafter, as shown in Fig. 6, the number in the sample code represents the heat number, and "a" and "c" represent the surface and center of the billet, respectively. Figure 6a shows the average diameters and densities of the inclusions in continuously cast billets. Overall, in the four heats, the average diameters of the inclusions on the surface of the billets were generally lower than those in the center of the billets, whereas the number densities of the inclusions on the surface of the billets were all greater than those in the center of the billets. As the rare earth element content in the steel sample increased from 0 to $142.22 \times 10^{-6}$, the average diameter of the inclusions on the surface of the billet generally decreased from 2.51 to 1.99 μm, that in the center of the billet decreased from 3.41 to 2.55 μm. In contrast, the inclusion density on the surface of the billet increased significantly from 20.5 to 51.8 $mm^{-2}$, and that in the center of the billet increased from 14.0 to 27.8 $mm^{-2}$. Figure 6b shows the size distributions of the total inclusions on the surface and in the center of the continuously cast billets. With the addition of rare earth elements from 0 to $25.67 \times 10^{-6}$ to $77.40 \times 10^{-6}$ and to $142.22 \times 10^{-6}$, the proportion of inclusions with a diameter of < 3 μm on the surface of the cast billet increased from 80.13% to 93.10% to 91.21% and to 90.28%; the proportion of inclusions with a diameter of 3–5 μm decreased from 11.54% to 4.02% to 8.20% and to 8.23%; and the proportion of inclusions with a diameter of < 3 μm in the center of the cast billet fluctuated slightly in the range of 76%–90%. Similarly, the proportion of inclusions with a diameter of 3–5 μm in the center of the cast billet also fluctuated in the range of 5%–18%. However, the general trend also indicated that after the addition of rare earth elements, the proportion of smaller inclusions increased, whereas that of larger inclusions decreased.

Figure 7 displays the compositional distributions of the inclusions in the continuously cast billets. Figure 7a, b shows the compositional distribution of inclusions in the

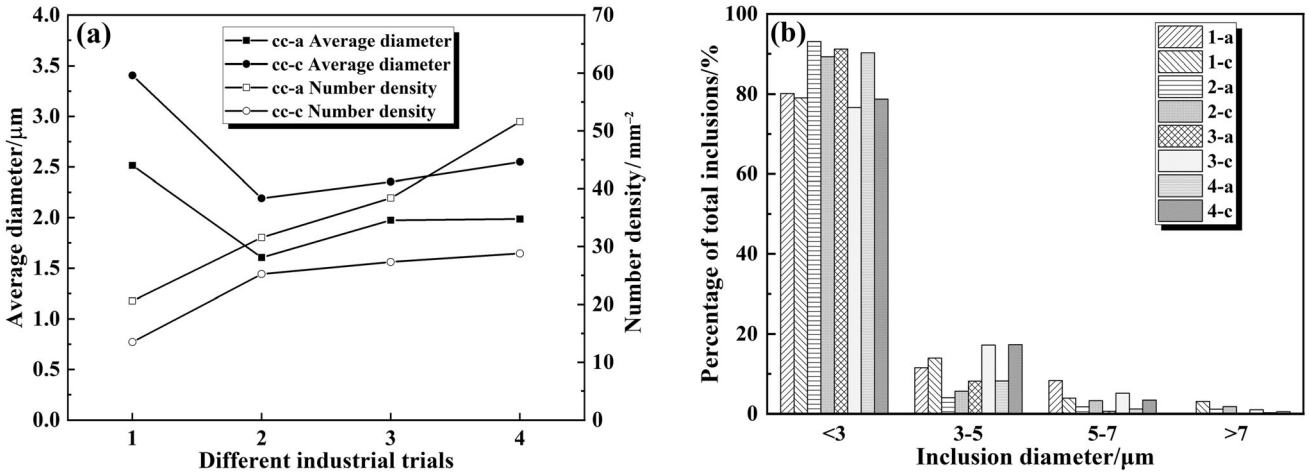



**Fig. 6** Changes in average diameter, number density (**a**) and size distribution (**b**) of total inclusions in continuously cast billets

continuous-casting billets of heat 1, indicating that without the addition of rare earth elements, the main inclusions in the continuous-casting billets were generally complex $CaO–SiO_2–Al_2O_3–MgO$ oxides with relatively low CaO contents (0–20 wt.%). Figure 7c–h shows the compositional distributions of the inclusions in CC-2, CC-3, and CC-4, respectively. The results indicated that in the continuously cast billets of heat 2, the main inclusions in the steel transformed from the $CaO–SiO_2–Al_2O_3–MgO$ phase to the $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases, with the content of $RE_2O_2S$ being the lowest among these components. In the continuous casting billets of heat 3, the main inclusions evolved into $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases, with the content of $REAlO_3$ being the lowest among these components. In the continuous casting billets of heat 4, the main inclusions were still $CaO–SiO_2–Al_2O_3–MgO$ + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases, with the content of $REAlO_3$ being the lowest among these components. According to the results shown in Fig. 7, the generation order of rare earth inclusions was $REAlO_3 \rightarrow RE_2O_3 \rightarrow RE_2O_2S$. In continuous casting billets, $REAlO_3$ and $RE_2O_3$ inclusions formed at relatively low REE contents. With increasing REE content, the amount of the $REAlO_3$ phase in the rare-earth phase gradually decreased, whereas the contents of the $RE_2O_2S$ and $RE_2O_3$ phases in the inclusions increased.

Figure 8 shows the average compositions of the total inclusions in the continuous casting billet. From heat 1 to 4, as the rare earth content in the casting billet increased from 0 to $25.67 \times 10^{-6}$ to $77.40 \times 10^{-6}$ and to $142.22 \times 10^{-6}$, the average rare earth inclusions content of the inclusions on the surface of the continuous casting billet gradually increased from 0 to 15.88%, 69.01%, and 70.14%, whereas the average rare earth inclusions content of the inclusions in the center of the continuous casting billet gradually increased from 0 to 8.75%, 64.38%, and 78.28%. An increase in the REE content in the steel clearly resulted in an increase in the rare earth content in the inclusions. Chemical compositions of the inclusions at the surface and in the center of the continuous casting billet were quite similar.

Figure 9 shows the typical inclusions in the continuous casting billets. Figure 9a, b displays the elemental distributions of typical inclusions in the continuous casting billet without the addition of rare earth elements. The two typical inclusions in the samples had the following compositions: $CaO–SiO_2–Al_2O_3–MgO–TiO_2–MnS$ and $CaO–SiO_2–Al_2O_3–MgO–TiO_2–MnO$. The $CaO–SiO_2–Al_2O_3–MgO–TiO_2–MnS$ inclusions generally had magnesium–aluminum spinel serving as the core, with the MnS precipitating on the surface. The $CaO–SiO_2–Al_2O_3–MgO–TiO_2–MnO$ inclusions were generally homogeneous and spherical. Figure 9c, d shows the typical inclusions in heat 2. The main inclusions in heat 2 were all generally composed of Mg–Al–Si–Ca–Ti–La–Ce–O–S and were irregular in shape. Some of the Mg–Al–Si–Ca–Ti–La–Ce–O–S inclusions had MnS phases precipitating on the surface. Figure 9e–h shows the typical Mg–Al–Si–Ca–Ti–La–Ce–O–S inclusions in heat 3 and heat 4, respectively. Most of them were homogeneous and spherical in shape in the two heats. In those steel samples, the inclusions with relatively high REE contents commonly presented white by SEM in BSE mode.

For the heat 1 before the rare earth treatment, the average diameter of inclusions increased from 2.12 μm in the LF sample to about 3 μm in the continuous casting billet, as well as the average number density of inclusions increased from 11.28 to about 17 $mm^{-2}$. Compared with that at the end of LF process, the proportion of inclusions with diameters larger than 3 μm in the continuous casting billet obviously increased, and the typical inclusions evolved from spherical homogeneous $CaO–SiO_2–Al_2O_3–MgO$ inclusions to $CaO–SiO_2–Al_2O_3–MgO–TiO_2–MnS$

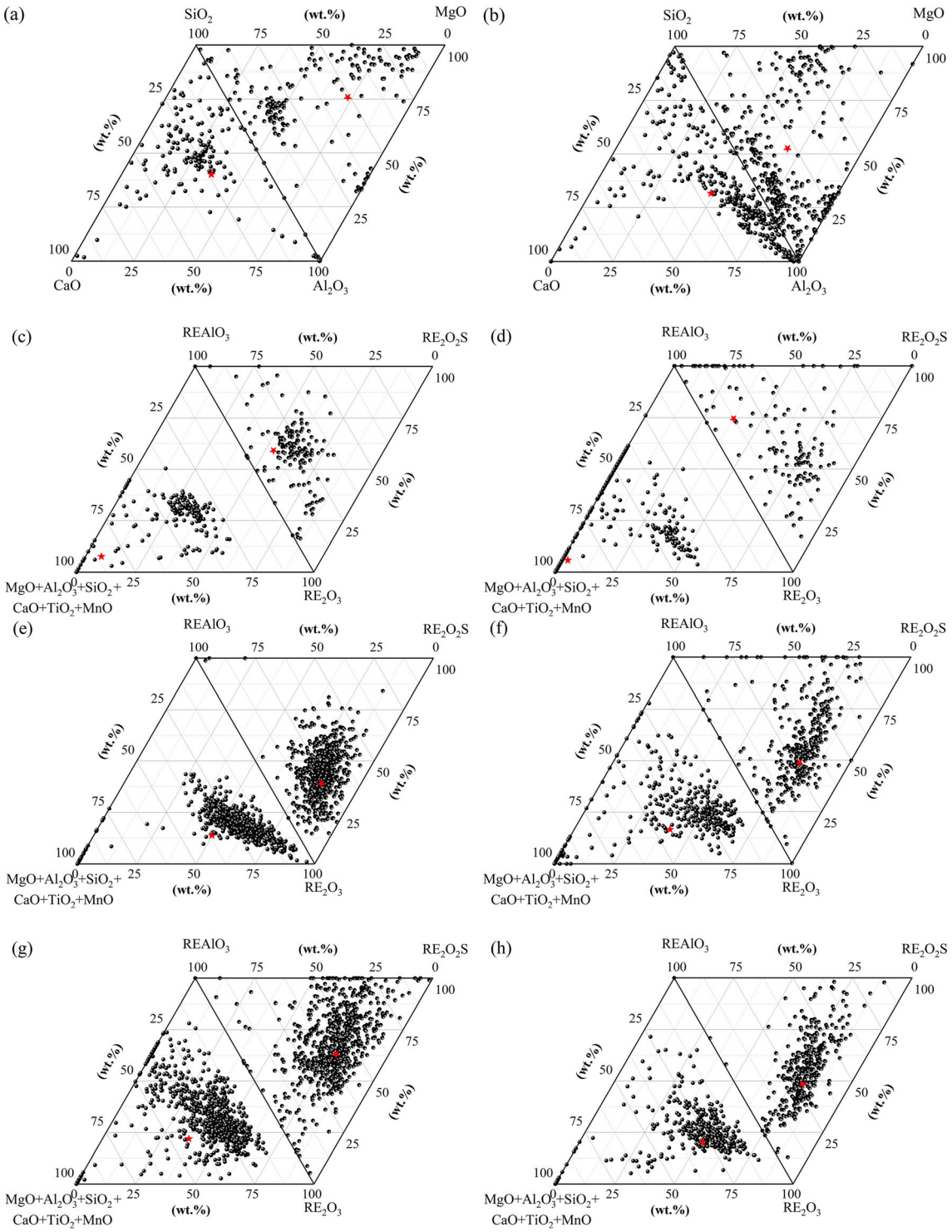



**Fig. 7** Distributions of inclusion compositions in continuous casting billets. **a** CC1-a; **b** CC1-c; **c** CC2-a; **d** CC2-c; **e** CC3-a; **f** CC3-c; **g** CC4-a; **h** CC4-c

inclusions due to the precipitation during cooling and solidification processes. After the rare earth treatment in the heat 1 to 3, the average diameter of inclusions increased from 1.3 μm in the LF sample to about 2.3 μm in the continuous casting billet, while the average number density of inclusions decreased from 70 to 35 $mm^{-2}$. The types and chemical compositions of typical inclusions in the LF samples and continuous casting billets after the rare earth treatment were almost the same which were all $CaO–SiO_2–Al_2O_3–MgO–TiO_2–REAlO_3–RE_2O_2S–RE_2O_3$, although

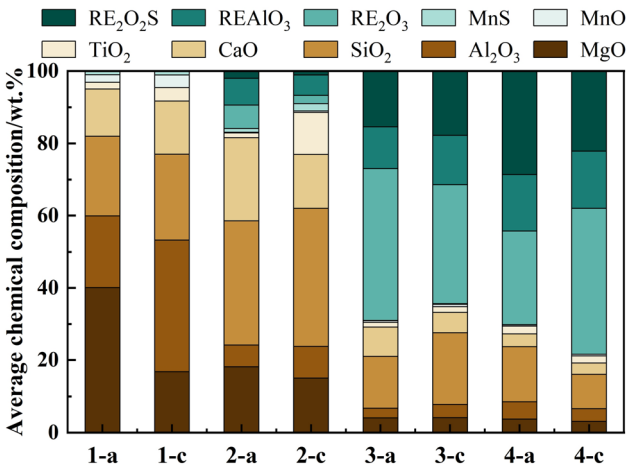



**Fig. 8** Average composition of inclusions in continuously casting billets

the rare earth element contents in the inclusions increased during cooling and solidification processes due to the shift of thermodynamic equilibrium between the steel and inclusions.

## 3.2 Corrosion resistance of steel before and after rare earth treatment

### 3.2.1 Electrochemical analysis

Figure 10 shows the electrochemical polarisation curves of the test steel with 3.5 wt.% NaCl solution, where $E$ and $i$ indicate the potential and the current density, respectively. Figure 10a displays the polarisation curves and Fig. 10b displays the anodic polarisation curves. All the samples were tested five times. The RE content had a significant effect on the electrochemical behaviour of 304 stainless steel in a neutral chloride solution. It could be observed from the curves that the stainless steel underwent oxygen reduction in the NaCl solution as the cathode reaction. Table 4 shows the corrosion current densities of 304 stainless steel with different RE contents in 3.5 wt.% NaCl solution which were derived using polarisation curves fitting. The corrosion current density of the steel was observed to be lower when rare earths were added, in comparison to the steel without the addition of rare earths. From heat 1 to heat 4, the corrosion current density of the

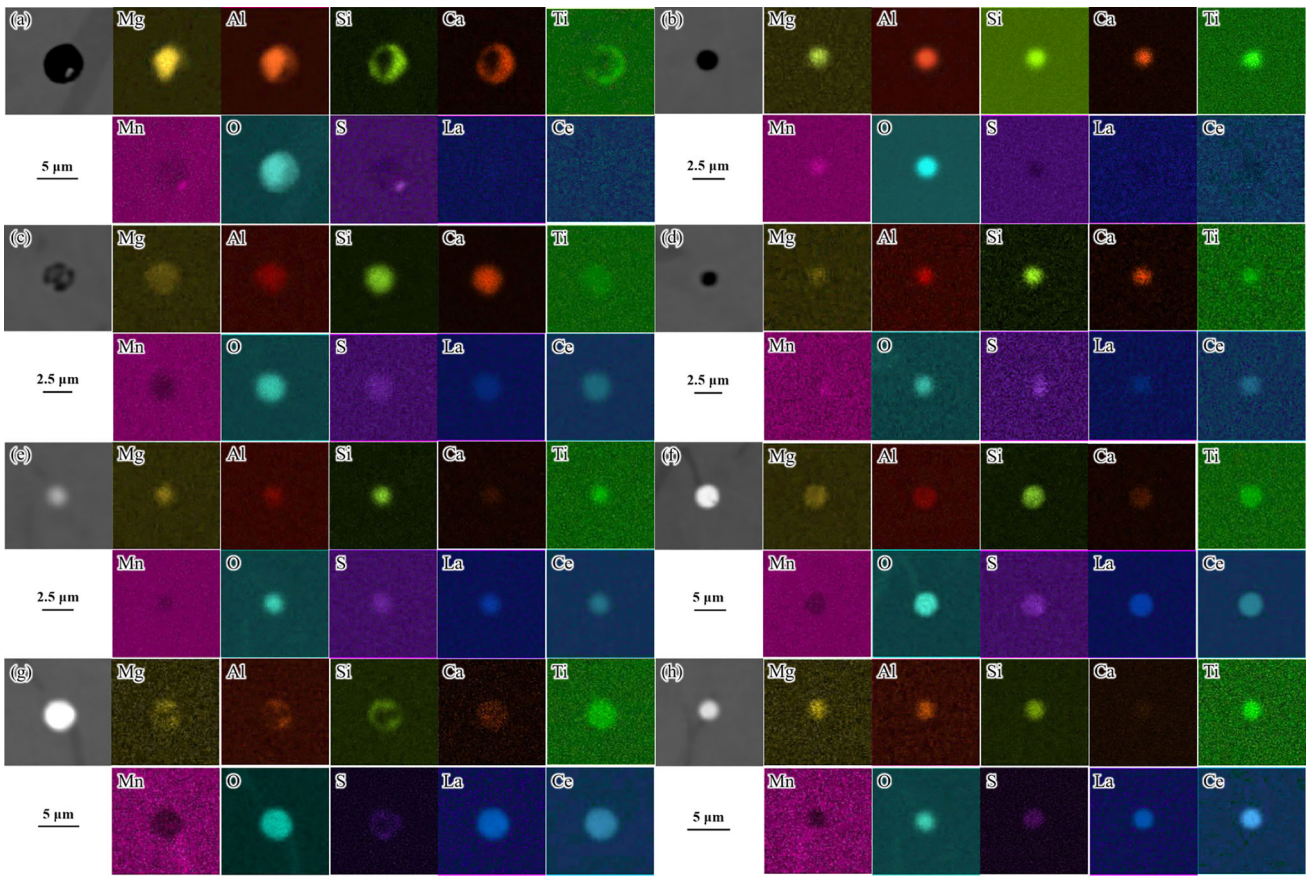



**Fig. 9** Morphologies and elemental distributions of typical inclusions in continuous casting billets of 304 stainless steel samples. **a**, **b** CC-1; **c**, **d** CC-2; **e**, **f** CC-3; **g**, **h** CC-4

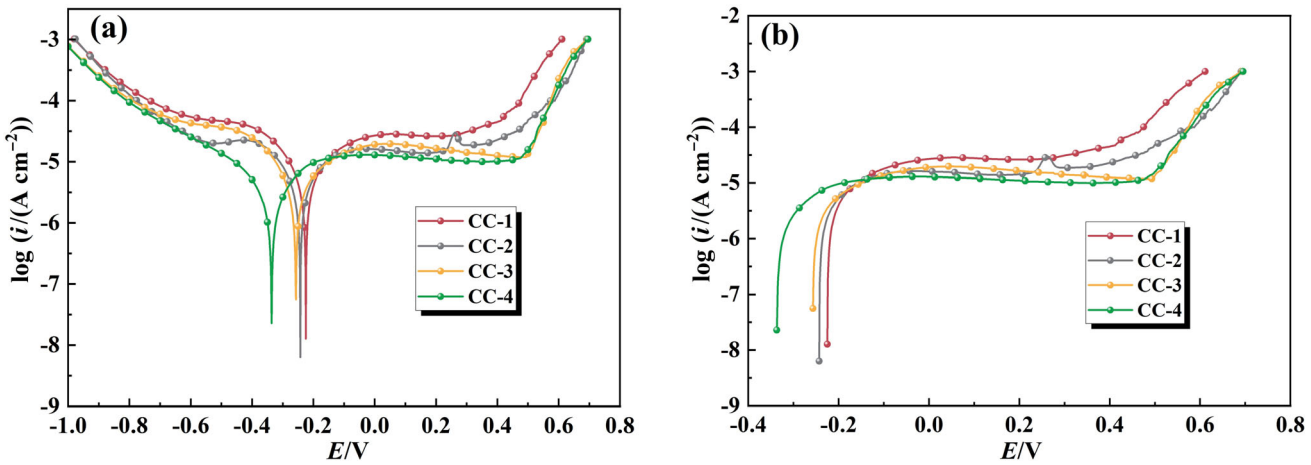



**Fig. 10** Potential polarisation curves of 304 stainless steel in 3.5 wt.% NaCl at room temperature. **a** Polarisation curve; **b** anodic polarisation curve

**Table 4** Corrosion current density of 304 stainless steel with different RE contents in 3.5 wt.% NaCl solution

| Serial number | Corrosion current density/ (μA cm$^{-2}$) |
|---|---|
| CC-1 | 7.253 |
| CC-2 | 6.593 |
| CC-3 | 5.439 |
| CC-4 | 3.412 |

cast billet samples decreased from 7.253 to 6.593 to 5.439 μA cm$^{-2}$ and finally to 3.412 μA cm$^{-2}$. It indicated that the corrosion resistance of the material improved with the increase in the rare earth content in the steel. Figure 10a shows that within the tested potential range, both the untreated sample and the samples with different rare earth contents displayed distinct passivation zones in the potential polarization curves. After rare earth treatment, the passivation zones of the samples notably widened, especially those with relatively high rare earth concentrations. This indicated that rare earth treatment enhanced the passivation properties of 304 stainless steel, and this enhancement was more pronounced with higher concentrations of rare earth elements within the experimental range. The anodic curves in Fig. 10b show that the steel with REE addition has a lower current, indicating that the anodic reaction on the steel surface was less intense. It was concluded that the corrosion resistance of the steel was improved by the addition of rare earths.

The Nyquist plots and dynamic potential polarization curves of samples with different rare earth element contents and untreated samples are presented in Fig. 11. Figure 11a–c shows Nyquist and Bode plots of 304 stainless steel with different RE contents, respectively. Figure 11d shows the equivalent circuit for quantitative evaluation of experimental impedance data. All the samples were tested five times. Nyquist plots were obtained from the electrochemical experimental data, accompanied by the corresponding fitting circuits. The fitting errors consistently remained below 10%. Additionally, owing to challenges in achieving ideal smoothness of electrode surfaces across all samples and ensuring consistent conductivity between the electrode surface adsorption layer and the detection solution, a "diffusion effect" was consistently observed. Consequently, the conventional capacitor C was replaced by a constant phase element (CPE) in the fitting circuit. In the equivalent circuit shown in Fig. 11d, $R_s$ is electrolyte resistance, and $R_1$ is passive film resistance. A larger passive film resistance indicated a higher corrosion resistance of the passive film on the material surface. Table 5 shows the fitting data. Postfitting analysis revealed that all the samples exhibited a single capacitive reactance arc. The size of the capacitive reactance arc was directly correlated with the resistance in the electrochemical reaction, indicating that a larger arc radius corresponded to enhanced corrosion resistance. The arc resistances of the samples with different rare earth contents were significantly greater than those of the untreated samples, as shown in Fig. 11a. Moreover, the radius of the capacitive arc and $R_1$ gradually increased with increasing rare earth content. This demonstrated that rare earth treatment effectively enhanced the corrosion resistance of 304 stainless steel. Furthermore, within the experimental concentration range, the corrosion resistance improved with increasing rare earth concentration, which was also consistent with the results obtained from the polarisation curves.

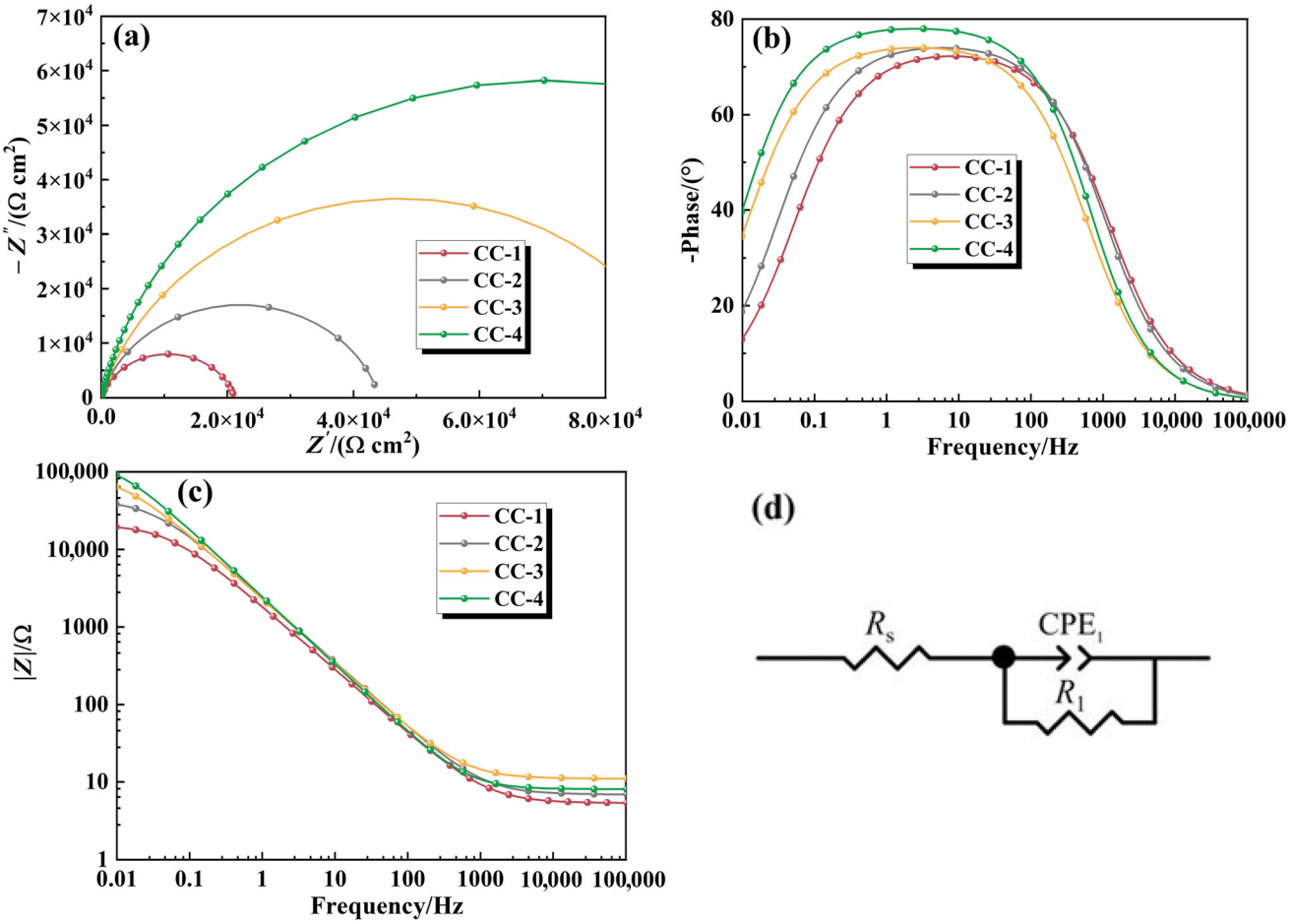



**Fig. 11** Nyquist (**a**) and Bode plots (**b**, **c**) of 304 stainless steel in 3.5 wt.% NaCl solution, and equivalent circuit (**d**) for quantitative evaluation of experimental impedance data. $Z'$—Real part of impedance; $Z''$—imaginary part of impedance; $|Z|$—modulus of impedance

**Table 5** Electrochemical fitting data of impedance spectra of 304 stainless steel with different RE contents in 3.5 wt.% NaCl solution

| Serial number | $R_s$/($\Omega$ cm$^2$) | $CPE_1$/($\Omega^{-1}$ cm$^{-2}$ s$^n$) | $n_1$/% | $R_1$/($\Omega$ cm$^2$) |
|---|---|---|---|---|
| CC-1 | 5.324 | 0.000119370 | 0.82174 | 21,166 |
| CC-2 | 6.863 | 0.000090292 | 0.83698 | 44,091 |
| CC-3 | 11.000 | 0.000094125 | 0.83544 | 94,860 |
| CC-4 | 8.050 | 0.000082093 | 0.87570 | 141,790 |

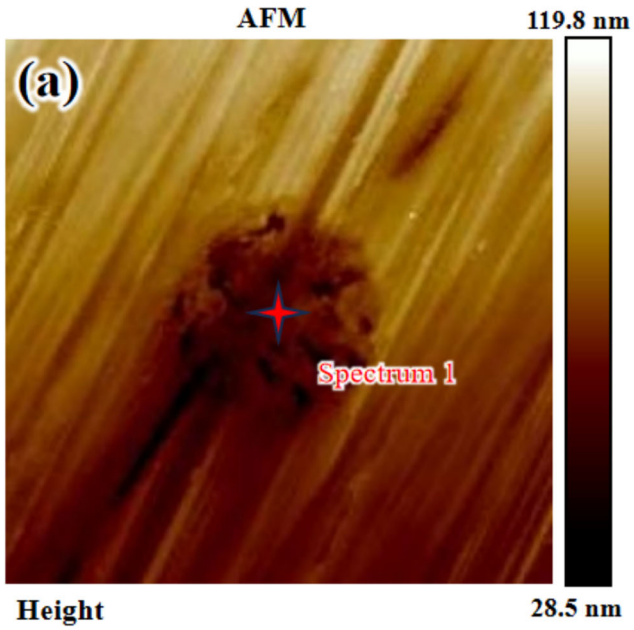
$n_1$—Fitting errors of $CPE_1$

### 3.2.2 Surface Volta potential

SKPFM was utilized to detect the Volta potential difference of typical inclusions to measure the localised electrochemical information in 304 stainless steel. SKPFM is commonly used to map Volta potential distributions on sample surfaces in air, which can represent the degree of electron escape from the inclusion and metal matrix at the sub-micron scale. Previous studies have reported that there was a close link between the corrosion tendency and Volta potential of metal materials, which was, the lower the Volta potential of a material, the higher corrosion tendency, and the easier it was to corrode the material in a corrosive environment. Therefore, the Volta potential could be used to predict the corrosion sensitivity of metallic materials [23–25].

After the immersion of the samples into 10 wt.% $FeCl_3$ solution for 10 h, the inclusions were located and analyzed via SKPFM. The composition of the inclusions was subsequently measured via energy dispersive spectroscopy (EDS). Figures 12–15 depict the surface morphology and potential difference, respectively, of four typical inclusions detected of CC-1, CC-2, CC-3 and CC-4 via AFM and SKPFM. Figures 12a, 13a, 14a and 15a show the height maps of four types of inclusions, revealing only slight scratches and a height difference of less than 100 nm. Therefore, the influence of height on SKPFM potential detection was assumed to be negligible.

Figure 12 shows that the typical inclusion in CC-1 was a Mg–Al–Si–Ca composite oxide without rare earth elements.




| Element | Spectrum 1 | |
|---|---|---|
| | wt.% | wt.% Sigma |
| C | 2.87 | 0.65 |
| O | 23.43 | 0.42 |
| Mg | 22.45 | 0.32 |
| Al | 1.61 | 0.11 |
| Si | 4.60 | 0.13 |
| S | 0.07 | 0.07 |
| Ca | 3.41 | 0.14 |
| Ti | 0.94 | 0.16 |
| Cr | 9.08 | 0.28 |
| Mn | 0.66 | 0.23 |
| Fe | 28.63 | 0.49 |
| Ni | 2.24 | 0.32 |
| La | 0.00 | 0.45 |
| Ce | 0.00 | 0.38 |

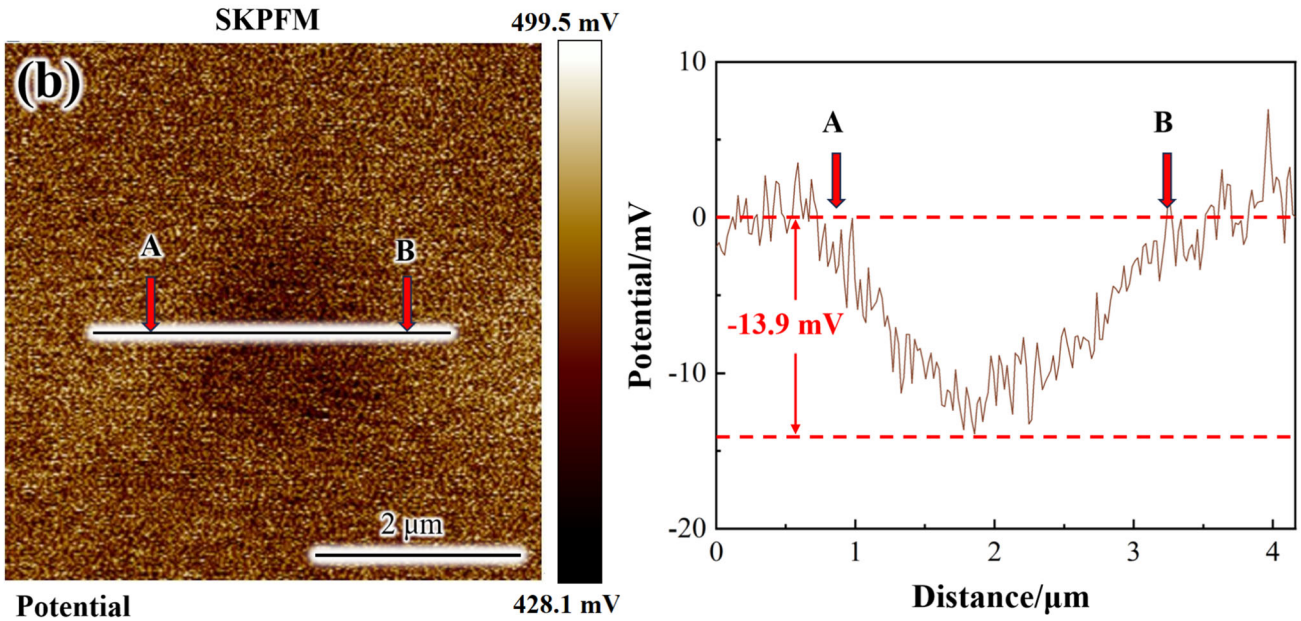
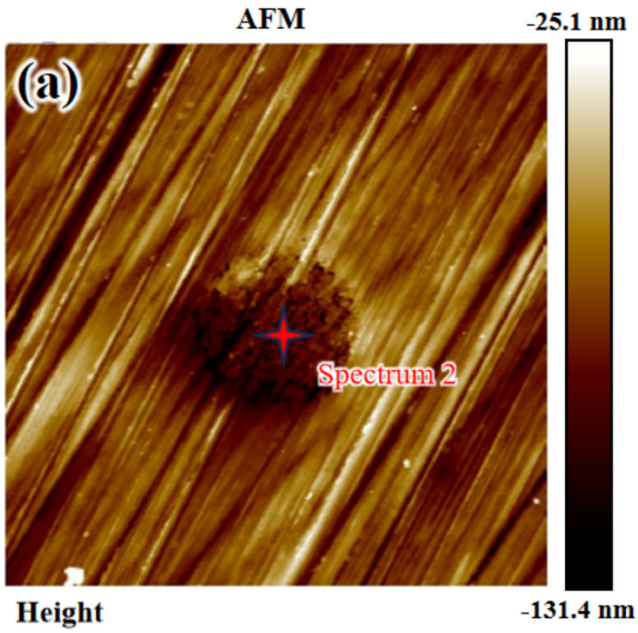



**Fig. 12** Surface morphologies and surface potentials around inclusions in CC-1. **a** Surface morphologies; **b** surface potentials and surface potential analyses. AFM—Atomic force microscope

Following immersion in the 10 wt.% $FeCl_3$ solution, the inclusion partially dissolved, with its height being slightly lower than that of the steel matrix surface. The SKPFM potential map demonstrated a considerable potential difference between the inclusion and the steel matrix, with a maximum potential difference of −13.9 mV, as shown in Fig. 12b. It indicated that the Mg–Al–Si–Ca composite oxide exhibited higher electrochemical activity than that of the steel matrix, predisposing them to localized dissolution and pitting corrosion. Figure 13 shows that the typical inclusion in CC-2 was a Mg–Al–Si–Ca composite oxide with small amounts of rare earth elements. After immersion in the 10 wt.% $FeCl_3$ solution, the inclusion also partially dissolved, with its height slightly lower than that of the steel matrix surface. Still, the SKPFM potential map showed a relatively great potential difference between the inclusion and the steel matrix, with a maximum potential difference of −12.5 mV. Similarly, this composite oxide also had a higher electrochemical activity than the steel matrix, predisposing it to localised dissolution and pitting corrosion.

EDS measurements demonstrated that the typical inclusions in CC-3 were REE-modified inclusions resulting




| Element | Spectrum 2 | |
|---|---|---|
| | wt.% | wt.% Sigma |
| C | 1.12 | 0.35 |
| O | 0.11 | 0.04 |
| Mg | 0.00 | 0.02 |
| Al | 0.03 | 0.03 |
| Si | 0.37 | 0.05 |
| S | 0.50 | 0.07 |
| Ca | 2.79 | 0.14 |
| Ti | 0.49 | 0.19 |
| Cr | 17.45 | 0.42 |
| Mn | 0.57 | 0.31 |
| Fe | 61.76 | 0.81 |
| Ni | 9.76 | 0.52 |
| La | 0.40 | 0.58 |
| Ce | 4.65 | 0.54 |

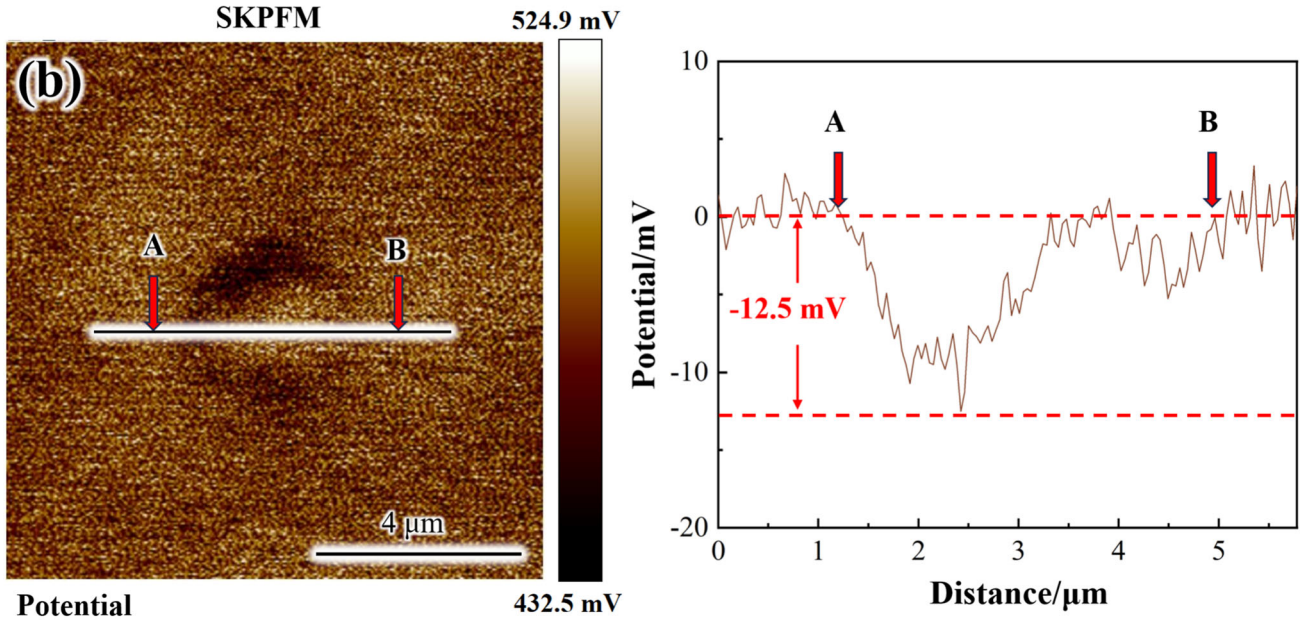
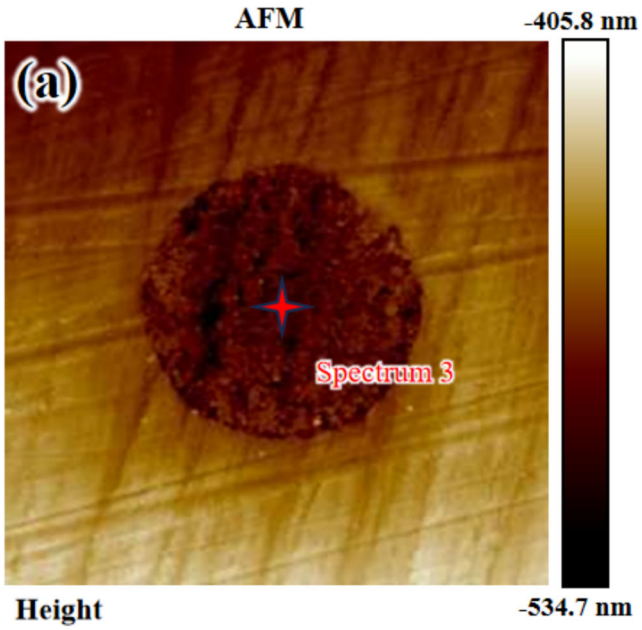



**Fig. 13** Surface morphologies and surface potentials around inclusions in CC-2. **a** Surface morphologies; **b** surface potentials and surface potential analyses

from the treatment with rare earth elements, as depicted in Fig. 14a. In comparison to those of the inclusions in CC-2 and CC-1 inclusions, the SKPFM potential map shown in Fig. 14b indicated a reduced potential difference between the inclusion and the steel substrate, reaching a maximum of −9.8 mV. Thus, the reduced electrochemical activity of the inclusions made them less susceptible to pitting in the corrosive anionic environment. The EDS-based compositions in Fig. 15a demonstrated the typical inclusion in CC-4 contained the highest rare-earth contents. The SKPFM potential map indicated that there was slight potential difference between the rare earth element-modified inclusions and the steel matrix. Consequently, the low electrochemical activity of the inclusions in CC-4 made them the least susceptible to pitting in aggressive anionic environments.

The aforementioned results demonstrated that the treatment of inclusions through rare earth modification could mitigate the potential difference between the inclusions and the steel matrix. This transformation converted inclusions with high electrochemical activity into those with low electrochemical activity, thereby enhancing the




| Element | Spectrum 3 | |
|---|---|---|
| | wt.% | wt.% Sigma |
| C | 3.50 | 0.67 |
| O | 14.28 | 0.42 |
| Mg | 1.35 | 0.13 |
| Al | 1.58 | 0.12 |
| Si | 4.52 | 0.15 |
| S | 1.07 | 0.10 |
| Ca | 1.95 | 0.12 |
| Ti | 1.25 | 0.23 |
| Cr | 6.15 | 0.32 |
| Mn | 0.54 | 0.25 |
| Fe | 20.99 | 0.50 |
| Ni | 2.58 | 0.34 |
| La | 17.69 | 0.74 |
| Ce | 22.55 | 0.70 |

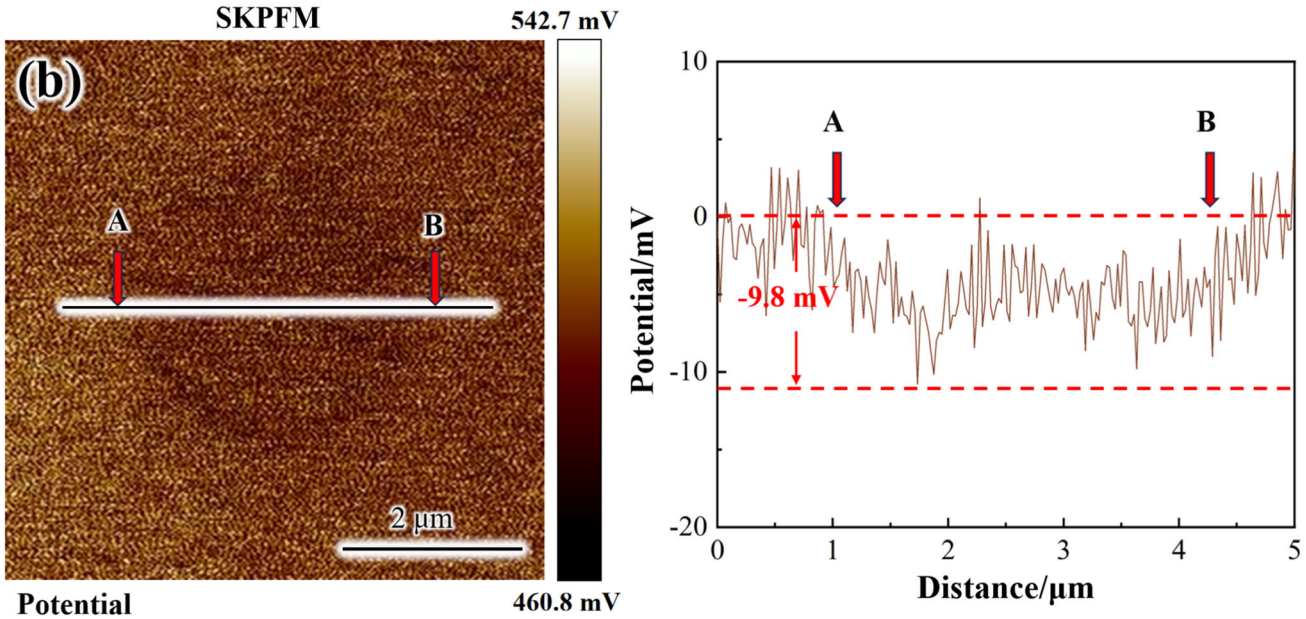
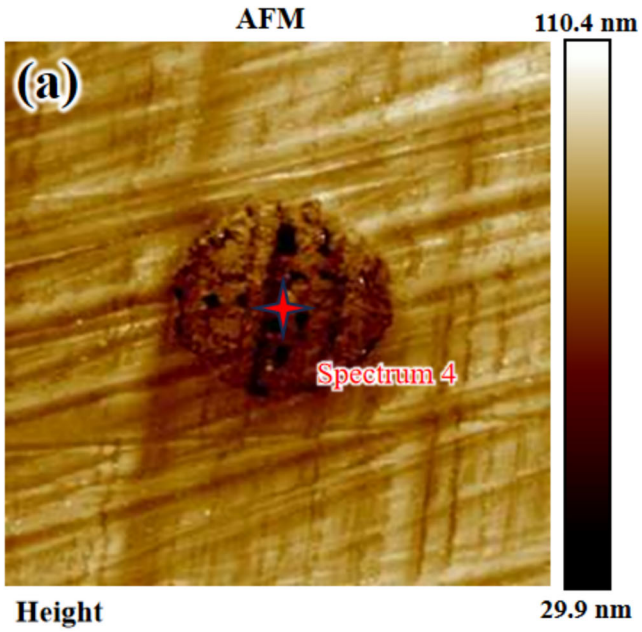



**Fig. 14** Surface morphologies and surface potentials around inclusions in CC-3. **a** Surface morphologies; **b** surface potentials and surface potential analyses

corrosion resistance of stainless steel. Furthermore, as the rare earth content increased, the potential difference between the inclusions and the steel matrix has been reduced.

### 3.2.3 Inclusion morphology

In order to investigate the form of corrosion caused by inclusions in each heat, a series of immersion tests were conducted. Figure 16 illustrates the SEM–EDS images of the morphology of typical inclusions from heat 1 to 4, both before and after corrosion for different times. These images were obtained from in situ observations before and after the corrosion of the inclusions. The orientation method was as follows: a cross was scratched at one corner of the sample, which served as the original point, and the coordinates of the inclusions were recorded and then the inclusion morphologies were observed in situ before and after corrosion. Figure 16a, c illustrates the typical inclusions observed in CC-1. Figure 16b, d illustrates the morphology of the corresponding inclusions following 50 h of immersion corrosion. These inclusions started pitting corrosion mainly




| Element | Spectrum 4 | |
|---|---|---|
| | wt.% | wt.% Sigma |
| C | 3.14 | 0.61 |
| O | 22.64 | 0.50 |
| Mg | 1.25 | 0.12 |
| Al | 1.69 | 0.12 |
| Si | 6.43 | 0.17 |
| S | 1.22 | 0.10 |
| Ca | 7.48 | 0.20 |
| Ti | 0.93 | 0.23 |
| Cr | 1.51 | 0.28 |
| Mn | 0.27 | 0.23 |
| Fe | 3.85 | 0.28 |
| Ni | 0.26 | 0.28 |
| La | 24.52 | 0.74 |
| Ce | 24.81 | 0.70 |

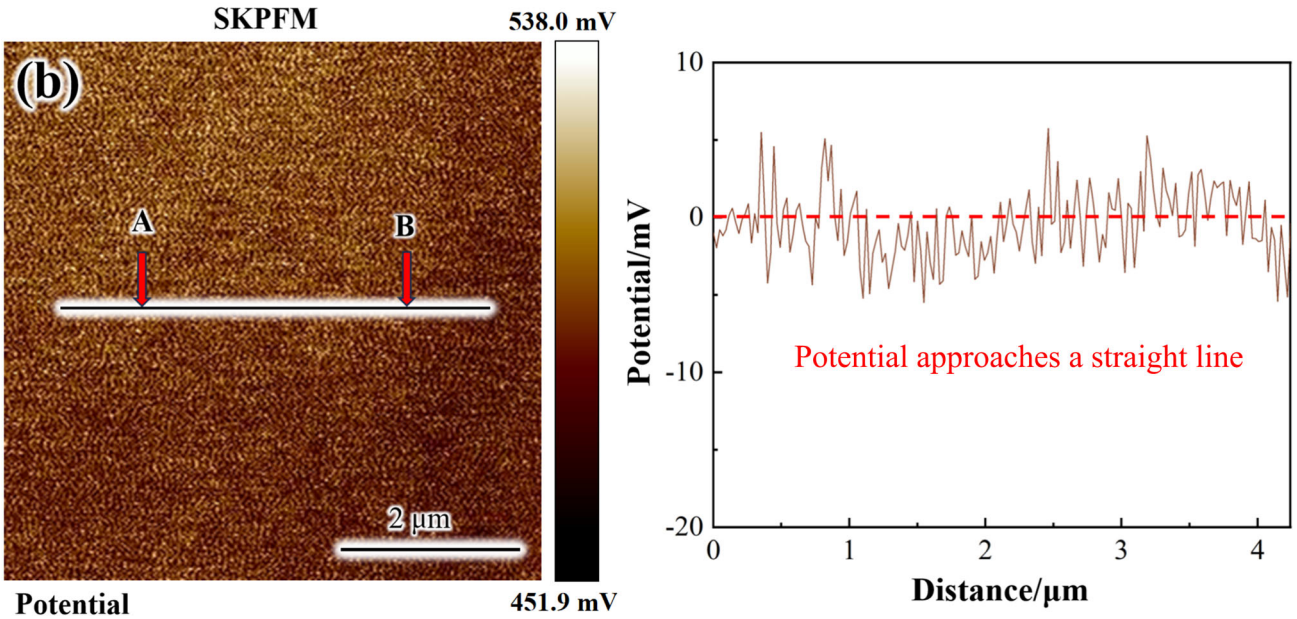



**Fig. 15** Surface morphologies and surface potentials around inclusions in CC-4. **a** Surface morphologies; **b** surface potentials and surface potential analyses

at MnS phase. After 50 h, a deep pit remained on the surface of the steel matrix. Figure 16e, g illustrates the typical inclusions observed in CC-2. Figure 16i, k illustrates the typical inclusions observed in CC-3. Figure 16m, o illustrates the typical inclusions observed in CC-4. With the increase of rare earth element content, the color of the inclusion gradually turned from black to white in the BSE mode of SEM. Figure 16f, h illustrates the morphology of the corresponding inclusions in CC-2 after 50 h of immersion corrosion, resulting in the formation of a deep pit on the surface of the steel matrix. Figure 16j, l illustrates the morphology of the corresponding inclusions in CC-3 after 50 h of immersion corrosion, while Fig. 16n, p depicts the morphology of the corresponding inclusions in CC-4 after the same period of immersion corrosion. After 50 h, a relative shallow pit formed and some inclusions remained on the surface of the steel substrate. It was consistent with previous SKPFM results which indicated that rare earth-modified inclusions reduced the severity of pitting corrosion. A higher rare earth content resulting in a greater reduction in severity.

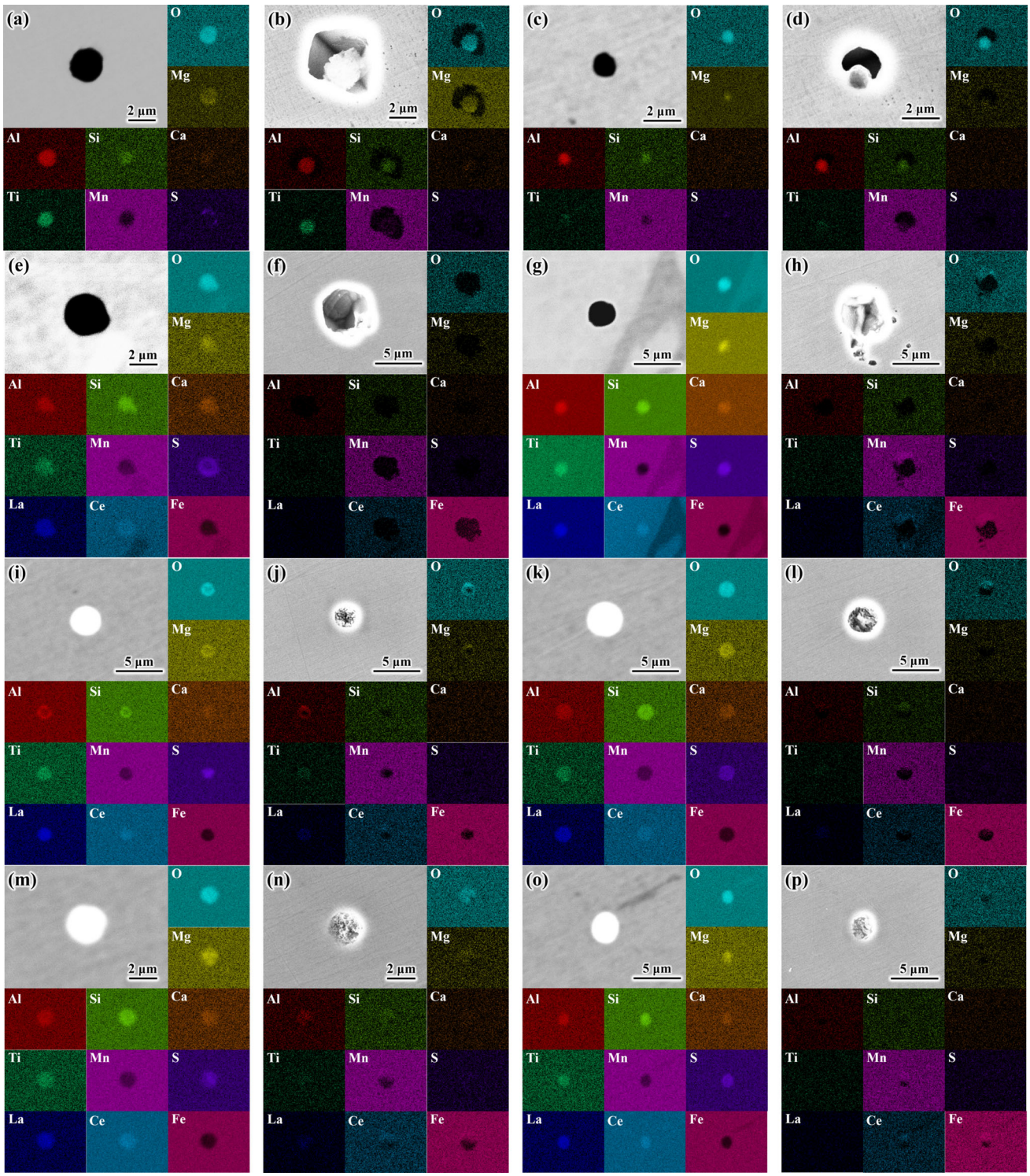



**Fig. 16** SEM–EDS images of localised corrosion morphology induced by different inclusions after immersion at different times. **a**, **c** CC-1 for 0 h; **b**, **d** CC-1 for 50 h; **e**, **g** CC-2 for 0 h; **f**, **h** CC-2 for 50 h; **i**, **k** CC-3 for 0 h; **j**, **l** CC-3 for 50 h; **m**, **o** CC-4 for 0 h; **n**, **p** CC-4 for 50 h

### 3.2.4 Localised corrosion mechanism

The mechanism of pitting initiated by the inclusions without rare earth elements differed from that of pitting initiated by the inclusions containing rare earth elements. The localized corrosion mechanisms of the MnS inclusions and Mg–Al–Si–Ca–Ti–O composite inclusions are illustrated in Fig. 17. For the MnS–Mg–Al–Si–Ca–Ti–O

inclusions, the MnS inclusions acted as the anodic phase and began to dissolve first (Fig. 17b). The dissolution of MnS resulted in the collection of corrosive ions in the pits, which created localized acidification and promoted further dissolution of the inclusions. As the corrosion process progressed, corrosion products accumulated in the pit, blocking the transfer of oxygen. Consequently, oxygen-concentration pools formed in the micropores, which subsequently accelerated the corrosion processes. As corrosion proceeded, MnS dissolved completely, leaving uncorroded Mg–Al–Si–Ca–Ti–O (Fig. 17c) on the steel matrix. Eventually, a deeper cavity remained on the surface of the steel matrix, and the presence of the cavity facilitated the ongoing hydrolysis reaction of the steel matrix.

The localized corrosion mechanisms of the Mg–Al–Si–Ca–Ti–La–Ce–S–O composite inclusions are shown schematically in Fig. 18. For these inclusions, localized corrosion was initiated when the inclusions were exposed to an aggressive solution (Fig. 18b). The dissolution of these inclusions produced corrosive ions. These ions accumulated in the corrosion pits and promoted further dissolution of the inclusions. Concurrently, $La^{3+}$ and $Ce^{3+}$ ions, generated by the corrosion of the inclusions, functioned as corrosion buffers that diminished anodic and cathodic activity, thereby impeding stainless steel corrosion. $Ce^{3+}$ and $La^{3+}$ within the pits impeded the sustained expansion of pitting corrosion and facilitated passivation [24]. During corrosion, inclusions with diminished rare earth content would undergo further corrosion, ultimately forming a substantial crater (Fig. 16f, h). Conversely, $La^{3+}$ and $Ce^{3+}$ derived from the initial corrosion of inclusions with elevated rare earth content would impede further corrosion, eventually halting the process and resulting in smaller pits (Fig. 16j, l, n, p).

## 4 Discussion

### 4.1 Thermodynamic calculations of changes in inclusion characteristics

Figure 19 shows the precipitates of inclusions during the cooling and solidification of steel calculated via FactSage 8.1. The steel composition used for the calculations was the same as that of LF-1. The REE content was altered to calculate the inclusion precipitation characteristics during the cooling and solidification of steels. The FactPS, FToxid and FTstel databases were used for the calculations. Owing to the similar properties of the REEs Ce and La and the widely recognized accuracies of thermodynamic calculations for Ce in molten steel [26–28], the content of La was converted to that of Ce to simplify the subsequent calculations.

For the calculation of Factsage, the contents of Mg, Al, Si, Ca, Ti, Mn, O, S, Cr, and Ni were the same as those of LF-1. From Fig. 19a–f, the rare earth contents increased in the following order: 0, $5 \times 10^{-6}$, $15 \times 10^{-6}$, $25 \times 10^{-6}$, $35 \times 10^{-6}$, and $75 \times 10^{-6}$. Figure 19a shows the calculated results for a rare earth content of 0. $Ca_2SiO_4$, MgO and CaO precipitated at 1873 K, and MnS precipitated at

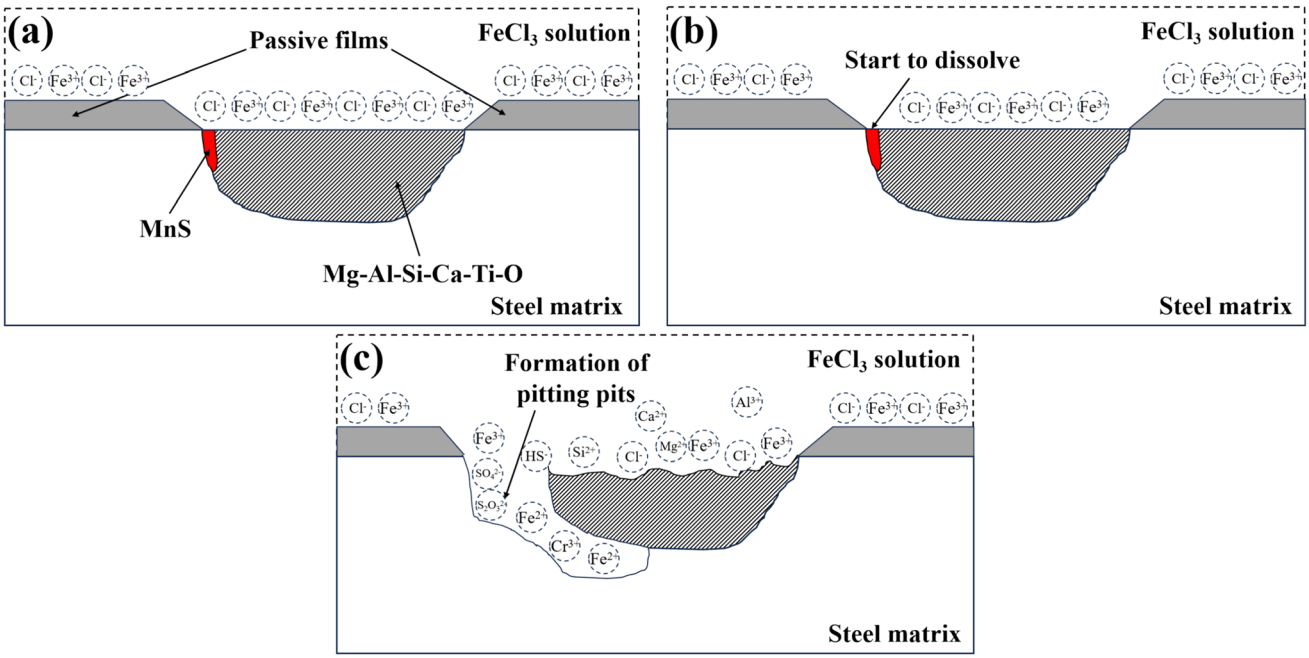



**Fig. 17** Schematic of pits in initiation and propagation induced by MnS–Mg–Al–Si–Ca–Ti–O inclusions

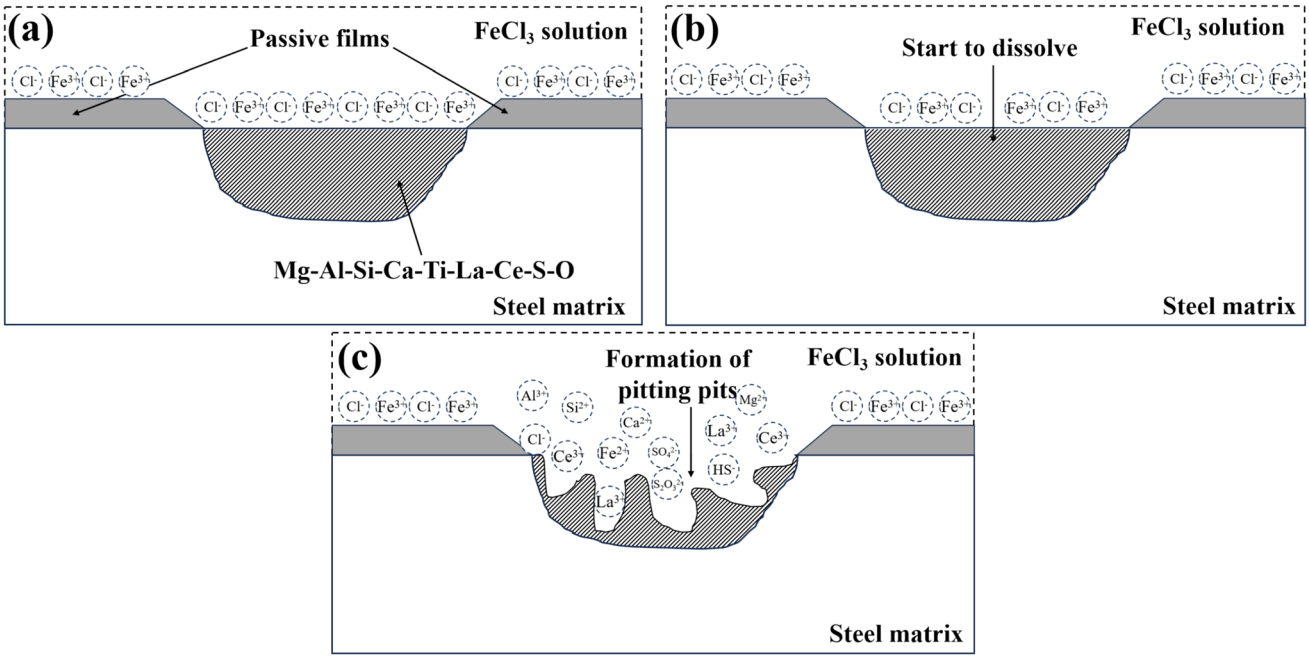



**Fig. 18** Schematic of pits in initiation and propagation induced by Mg–Al–Si–Ca–Ti–La–Ce–S–O inclusions

lower temperatures. Figure 19b–d shows the results of calculations for rare earth contents of $5 \times 10^{-6}$, $15 \times 10^{-6}$, and $25 \times 10^{-6}$, respectively. $Ca_2SiO_4$, MgO, CaO, and $Ce_2O_3$ precipitated at 1873 K, $CeAlO_3$ precipitated at almost 1773 K, and MnS precipitated at a lower temperature. Figure 19e, f shows the results of the calculations for rare earth contents of $35 \times 10^{-6}$ and $75 \times 10^{-6}$. $Ca_2SiO_4$, MgO, CaO, and $Ce_2O_3$ precipitated at 1873 K, and $CeAlO_3$ and $Ce_2O_2S$ precipitated at lower temperatures. Figure 19 shows that the REE contents in the inclusions increased with increasing REE contents in the steels. The calculations also show that with increasing REE content, the REE inclusions formed in the order of $CeAlO_3 + Ce_2O_3 \rightarrow CeAlO_3 + Ce_2O_3 + Ce_2O_2S$. $Ce_2O_3$ precipitated at relatively high temperatures, while $CeAlO_3$ and $Ce_2O_2S$ precipitated when the temperature decreased, and $Ce_2O_2S$ started to appear at relatively high REE contents.

## 4.2 Mechanism analyses of changes in inclusion characteristics

Given the similar properties of the REEs Ce and La and the widely recognized accuracy of thermodynamic calculations for Ce in molten steel, the content of La was also converted to that of Ce to simplify the subsequent calculations.

The main inclusion in steels without additional rare earth elements was Mg–Al–Si–Ca–O. With the addition of rare earth elements, $CeAlO_3$, $Ce_2O_3$ and $Ce_2O_2S$ formed in the steel due to the strong affinity of Ce for Al, O and S. The standard Gibbs free energy variations ($\Delta G^{\circ}$) for the formation of these inclusions are shown in Eqs. (1)–(3). When the equation reaches equilibrium, the Gibbs free energy is equal to zero, and Eq. (4) can be obtained.

$$[Ce] + [Al] + 3[O] = (CeAlO_3) \quad \Delta G^{\circ} = -1,366,460 + 364T \tag{1}$$

$$2[Ce] + 3[O] = (Ce_2O_3) \quad \Delta G^{\circ} = -1,431,090 + 360.06T \tag{2}$$

$$2[Ce] + 2[O] + [S] = (Ce_2O_2S) \quad \Delta G^{\circ} = -1,353,592.4 + 331.6T \tag{3}$$

$$\ln K = -\Delta G^{\circ}/(RT) \tag{4}$$

where $K$ is the equilibrium constant; $R$ is the ideal gas constant; and $T$ is the temperature.

Equation (5) is for the precipitation of MnS in the liquid steel. Combining Eqs. (3) and (5) leads to Eqs. (6) and (7), which are the chemical reactions for Ce-modified MnS. According to the data in Fig. 5, the unmodified inclusions contained a portion of $Al_2O_3$ and a small amount of sulphide. During the LF refining process at 1873 K, Ce element would react with $Al_2O_3$ component to form $REAlO_3$. Besides, the calculation results in Eq. (7) showed that when $G = 0$, $a^2_{[Ce]}a_{MnS} = 6.17033^{-17}$. It indicated that even at very low concentration of Ce element, MnS phase in the inclusions would be modified rapidly by rare earth treatment to generate $RE_2O_2S$. With the increase in Ce content, $RE_2O_2S$ would transform to $RE_2O_3$ due to the further reaction between Ce and O element. Whereas for continuous casting billets, inclusions reacted with the steel for a

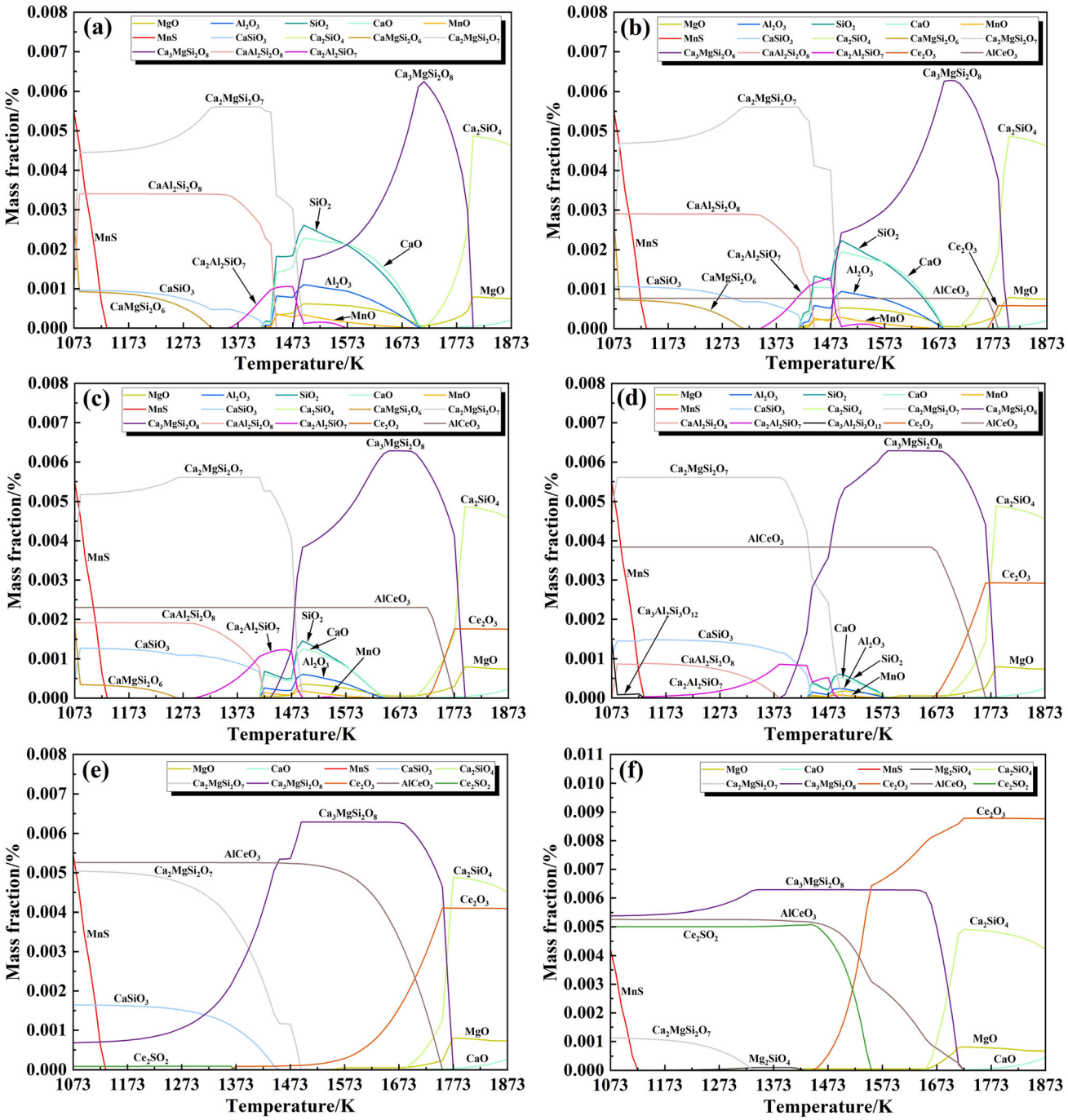



**Fig. 19** Equilibrium cooling phase diagrams of 304 stainless steels with different rare earth element contents. **a** 0 Ce; **b** $5 \times 10^{-6}$ Ce; **c** $15 \times 10^{-6}$ Ce; **d** $25 \times 10^{-6}$ Ce; **e** $35 \times 10^{-6}$ Ce; **f** $75 \times 10^{-6}$ Ce

relatively long time, the composition of the inclusions should be close to the equilibrium composition during cooling and solidification processes. Therefore, in the continuous casting process, the generation order of rare earth inclusions was $REAlO_3 \rightarrow RE_2O_3 \rightarrow RE_2O_2S$, according to the thermodynamic calculations of the changes in inclusion characteristics, as shown in Fig. 19. The active interaction coefficients used in this study are listed in Table 6 [29, 30].

$$[\mathrm{Mn}] + [\mathrm{S}] = \mathrm{MnS} \quad \Delta G^{\circ} = -168,822 + 98.87T \tag{5}$$

$$2[\mathrm{Ce}] + 2[\mathrm{O}] + \mathrm{MnS} = \mathrm{Ce_2O_2S} + [\mathrm{Mn}] \quad \Delta G^{\circ} = -118,770.4 + 232.73T \tag{6}$$

$$\Delta G = \Delta G^{\circ} + RT\ln K = -118,770.4 + 232.73T + 8.314T\ln\frac{a_{[Mn]}a_{Ce_2O_2S}}{a_{[Ce]}^2 a_{[O]}^2 a_{MnS}} \quad (7)$$

where $a_{[i]}$ is the activity of element [$i$] in molten steel; and $a_{M_xO_y}$ and $a_{MnS}$ are the activity of the oxide $M_xO_y$ and MnS in the inclusions, respectively.

By combining the test results, thermodynamic calculations, and theories of other scholars on rare-earth-modified inclusions [31–34], the transformation mechanisms of the characteristics of the inclusions in the 304 stainless steel samples were proposed and shown in Fig. 20.

The first was the transformation of inclusions in the LF refining process. Without the addition of rare earth elements, the main inclusions in the steels after LF refining were generally complex CaO–$SiO_2$–$Al_2O_3$–MgO oxides. As the content of rare earth elements (Ce + La) increased from 0 to $2.2 \times 10^{-6}$, the main inclusions in the steel after LF refining gradually transformed from the CaO–$SiO_2$–$Al_2O_3$–MgO phase to the CaO–$SiO_2$–$Al_2O_3$–MgO + ($REAlO_3$ and $RE_2O_2S$) phases. As the content of rare earth elements (Ce + La) increased to $25.2 \times 10^{-6}$ in the LF-3 steel sample, the main inclusions evolved to CaO–$SiO_2$–$Al_2O_3$–MgO + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases. When the content of rare earth elements (Ce + La) reached $50.37 \times 10^{-6}$ in the LF-4 steel sample, the main inclusions were still CaO–$SiO_2$–$Al_2O_3$–MgO + ($REAlO_3$, $RE_2O_2S$ and $RE_2O_3$) phases.

The second aspect was the evolution of inclusions in the tundish process from different heats. In the samples without added rare earth elements, there were two main types of inclusions, $SiO_2$–$Al_2O_3$–$TiO_2$–MnO–MnS and CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$. The $SiO_2$–$Al_2O_3$–$TiO_2$–MnO–MnS inclusions were homogeneous in composition and spherical in shape. The diameter of this type of inclusion was small, which formed by combining Mg, Al and other elements after the oxygen entering the steel. The CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$ inclusions were generally spherical, with Mg–Al–O serving as the core. In addition, CaO–$SiO_2$–$TiO_2$ wrapped around the exteriors of the inclusions. This type of inclusion had a large diameter and evolved from the original inclusions in the steel. The inclusions of the samples with REE contents of $25.67 \times 10^{-6}$ were composed of CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–$REAlO_3$–$RE_2O_2S$–$RE_2O_3$. In these inclusions, Mg, Al, Si, Ca, and Ti were concentrated in the centers of the inclusions, while the distributions of S, La, and Ce in the inclusions were relatively homogeneous. In this sample, some inclusions had high REE contents and appeared white in the BSE mode of SEM. The main inclusions of the samples with REE contents of $77.40 \times 10^{-6}$ and $142.22 \times 10^{-6}$ were all spherical CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–$REAlO_3$–$RE_2O_2S$–$RE_2O_3$ inclusions containing very little Mg, Al, Si, or Ca contents.

The third aspect is the evolution of inclusions in continuously cast billets from different heats. Without the addition of rare earth elements, the main inclusions in the steels in the continuously cast billets were complex CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–MnS and CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–MnO. The CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–MnS inclusions generally had magnesium–aluminum spinel serving as the core and MnS phase precipitating on the surface. The CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–MnO inclusions were generally homogeneous and spherical. The main inclusions in heat 2 were generally CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–$REAlO_3$–$RE_2O_2S$–$RE_2O_3$, which were irregular in shape. Some of them had MnS precipitating on the surface. The main inclusions in heat 3 and heat 4 were still generally composed of the CaO–$SiO_2$–$Al_2O_3$–MgO–$TiO_2$–$REAlO_3$–$RE_2O_2S$–$RE_2O_3$ phase. Some of them had $RE_2O_2S$-$RE_2O_3$ precipitating on the surface in heat 3, whereas others displayed $TiO_2$–$RE_2O_2S$–$RE_2O_3$ as the core in heat 4. The inclusions in the continuous casting billet were further developed on the basis of those in the tundish process, which were relatively similar in morphology and composition.

### 4.3 Analysis on ability of rare earth elements to improve pitting resistance

In this study, the addition of rare earth elements to stainless steel enhanced the corrosion resistance of the steel. Drawing on the findings of other scholars [35, 36], this enhancement could be explained on the basis of three factors. First, rare earth elements could modify the original

**Table 6** First-order interaction coefficients for thermodynamic calculations

| Element | Si | Mn | Ca | S | Al | Ti | O | Ce | Cr | Ni |
|---|---|---|---|---|---|---|---|---|---|---|
| Al | 0.056 | 0.035 | −0.047 | 0.03 | 0.045 | | −6.6 | −0.43 | 0.025 | 0.008 |
| O | −0.131 | −0.021 | −271 | −0.133 | −3.9 | −0.6 | −0.2 | −0.57 | −0.04 | 0.006 |
| S | 0.063 | −0.026 | −110 | −0.028 | 0.035 | 0.072 | −0.27 | −1.88 | −0.011 | −0.064 |
| Ce | – | 0.13 | – | −8.39 | −2.25 | −3.62 | −5.03 | −0.0039 | – | – |

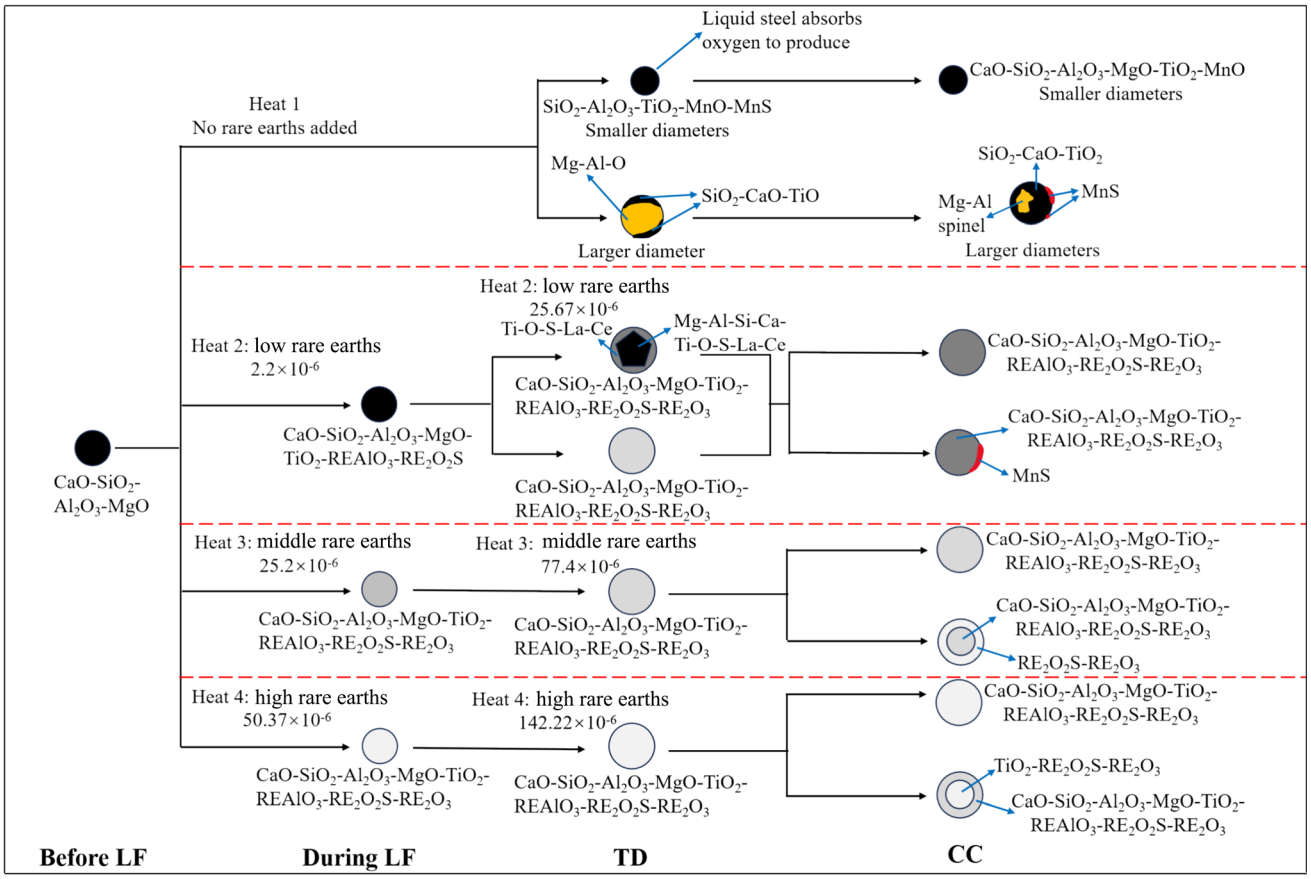



**Fig. 20** Schematic diagram of inclusion changes during industrial production of 304 stainless steel. TD—Tundish; CC—continuous casting

$CaO–SiO_2–Al_2O_3–MgO$ inclusions in steel into more stable rare earth inclusions. The modified inclusions themselves exhibited good corrosion resistance and did not dissolve first during pitting. The $CaO–SiO_2–Al_2O_3–MgO$ inclusions without rare earth elements had a significant potential difference from the steel substrate, with a maximum potential difference of −13.9 mV, as shown in Fig. 12. In contrast, inclusions with rare earth elements did not exhibit such a significant potential difference from the steel substrate. The results indicated that REE-modified inclusions had a lower voltage potential difference than unmodified $CaO–SiO_2–Al_2O_3–MgO$ inclusions did. This finding suggested that REE-modified inclusions were less likely to cause galvanic coupling corrosion. Second, the mechanism of pitting initiated by inclusions without rare earth elements differed from that of pitting initiated by inclusions containing rare earth elements. For the MnS–Mg–Al–Si–Ca–Ti–O inclusions, the MnS inclusions acted as the anodic phase and began to dissolve first. The dissolution of MnS resulted in the collection of corrosive ions in the pits, which created localized acidification and promoted further dissolution of the inclusions. As the corrosion process progressed, corrosion products accumulated in the pit, blocking the transfer of oxygen. Consequently, oxygen-concentration pools formed in the micropores, which subsequently accelerated the corrosion process. As corrosion proceeded, MnS dissolved completely, leaving uncorroded Mg–Al–Si–Ca–Ti–O on the steel matrix. Eventually, a deeper cavity remained on the surface of the steel matrix, and the presence of the cavity facilitated the ongoing hydrolysis reaction of the steel matrix. For the Mg–Al–Si-Ca-Ti–La–Ce–S–O inclusions, localized corrosion was initiated when the inclusions were exposed to an aggressive solution. Concurrently, $La^{3+}$ and $Ce^{3+}$ ions, generated by the corrosion of the inclusions, functioned as corrosion buffers that diminish anodic and cathodic activity, thereby impeding stainless steel corrosion. $Ce^{3+}$ and $La^{3+}$ within the pits impeded the sustained expansion of pitting corrosion and facilitated passivation. During corrosion, inclusions with lower rare earth content would undergo further corrosion, ultimately forming a substantial crater. Conversely, $La^{3+}$ and $Ce^{3+}$ derived from the initial corrosion of inclusions with elevated rare earth content would impede further corrosion, eventually halting the process and resulting in smaller pits. Third, the addition of rare earth elements to stainless steel significantly reduced the number of large inclusions in the steel. Small inclusions took longer to grow as pitting nucleation sites and were less

likely to form stable pitting pits. In this study, the diameters of the inclusions in the continuously cast billets decreased after the addition of rare earth elements. With the addition of rare earth elements from 0 to $25.67 \times 10^{-6}$ to $77.40 \times 10^{-6}$ and to $142.22 \times 10^{-6}$, the proportion of inclusions with a diameter of $< 3$ μm on the surface of the cast billet increased from 80.13% to 93.10% to 91.21% and to 90.28%. In contrast, the proportion of inclusions with a diameter of 3–5 μm decreased from 11.54% to 4.02% to 8.20% and to 8.23%, while the proportion of inclusions with a diameter of $< 3$ μm in the center of the cast billet slightly fluctuated in the range of 76%–90%. Similarly, the proportion of inclusions with a diameter of 3–5 μm in the center of the cast billet also fluctuated in the range of 5%–18%. However, the general trend also indicated that after the addition of rare earth elements, the proportion of smaller inclusions increased, whereas that of larger inclusions decreased. In summary, the reduced potential difference between the modified inclusions and the steel substrate and the refined inclusions decreased the probability of pitting corrosion initiated by inclusions, minimized the risk of galvanic coupling corrosion, and generally improved the corrosion resistance of 304 stainless steel.

## 5 Conclusions

1. In the LF refining process, with increasing rare earth content from 0 to $50.37 \times 10^{-6}$, REE inclusions were preferentially generated in the following order when they were modified: $REAlO_3 \rightarrow RE_2O_2S \rightarrow RE_2O_3$. With increasing REE content, the amount of the $REAlO_3$ phase gradually decreased, whereas the contents of the $RE_2O_2S$ and $RE_2O_3$ phases in the inclusions increased.
2. The inclusion sizes in the continuously cast billets obviously decreased after rare earth elements were added, whereas the inclusion densities increased with increasing REE contents. In addition, the formation of Mg–Al spinel and MnS in the continuously cast billet was suppressed by the addition of rare earth elements.
3. Increasing the rare earth content improved the corrosion resistance of 304 stainless steel. The voltage potential differences between the REE-modified inclusions and the steel substrate were obviously less than those between the unmodified inclusions and the steel substrate, which decreased the probability of pitting corrosion initiated by inclusions and minimized the risk of galvanic coupling corrosion.
4. The incorporation of rare earth elements alters the mechanism of inclusions-induced pitting. The third and fourth heat inclusions exhibited a higher concentration of rare earth elements, thereby markedly reducing pitting induced by the inclusions.

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## Declarations

**Conflict of interest** Cheng-song Liu is a youth editorial board member for *Journal of Iron and Steel Research International* and was not involved in the editorial review or the decision to publish this article. On behalf of all authors, the corresponding author states that there is no conflict of interest.

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